1 Systems of Linear Equations

1.1 Forming the Equations

We read the context of the question and form equations depending on the context of the question.

Example 1.1

$$\left.\begin{aligned} a+b+c&=176 && \text{—(1)} \\ 0.85a+0.90b+0.95c&=161.80 && \text{—(2)} \\ 0.93a+0.95b+0.98c&=169.56 && \text{—(3)} \end{aligned}\right\} \Rightarrow a=32,\ b=44,\ c=100$$

Example 1.2

The “variables” may be expressions. To find x, y and z, use the given information to form the equations:

$$\left.\begin{aligned} 2x^3+3\ln y-\mathrm{e}^z&=18 && \text{—(1)} \\ x^3-\ln y+2\mathrm{e}^z&=9 && \text{—(2)} \\ x^3+2\ln y+\mathrm{e}^z&=11 && \text{—(3)} \end{aligned}\right\} \Rightarrow x^3=8,\ \ln y=1,\ \mathrm{e}^z=1$$

It is technically still ‘linear’ because we can think of $A=x^3$, $B=\ln y$, $C=\mathrm{e}^z$, which turns it into a linear system in A, B and C. A truly nonlinear equation could look like $x^2+x+y^2+2y-3z^3-xz+xy=1$; it cannot be made linear this way, because x appears as both x^2 and x, and there are product terms such as xz and xy.

1.2 Special Case: Number of Equations > Number of Variables

Sometimes the number of equations is more than the number of variables, and the solution must satisfy all of the equations.

Example 1.3

$$\left.\begin{aligned} \lambda+\mu&=1 && \text{—(1)} \\ 2\lambda-\mu&=5 && \text{—(2)} \end{aligned}\right\} \Rightarrow \lambda=2,\ \mu=-1$$

$$\lambda+3\mu=-1 \qquad \text{—(3)}$$

Sometimes there may be no solution, e.g. skew lines in vectors. In this case, solve for λ and μ using equations (1) and (2). Then sub into (3).

Note

Check (3): if LHS = RHS, the equations are consistent; if LHS ≠ RHS, there is no solution.

1.3 Special Case: Number of Equations < Number of Variables

Sometimes the number of equations is less than the number of variables, so there are infinitely many solutions.

Example 1.4

$$\left.\begin{aligned} x+y+z=6 \quad &\text{—(1)} \\ x-y+2z=5 \quad &\text{—(2)} \end{aligned}\right\} \Rightarrow x=\tfrac{11}{2}-\tfrac{3}{2}z,\ y=\tfrac{1}{2}+\tfrac{1}{2}z,\ z=z$$

Note

Useful trick: in contextual questions, the variables are usually non-negative integers, $z \in \mathbb{Z}_0^+$, so use this to find the solution from the general solution.

When $z=z$, it means that there are infinitely many solutions, and z can be any real value. So just sub that into x and y to get a correct solution.

$z=-1:\ (7,0,-1)$ $z=0:\ (\frac{11}{2},\frac{1}{2},0)$ $z=1:\ (4,1,1)$ $z=2:\ (\frac{5}{2},\frac{3}{2},2)$

$z=3:\ (1,2,3)$ $z=4:\ (-\frac{1}{2},\frac{5}{2},4)$ $z=5:\ (-2,3,5)$ $z=6:\ (-\frac{7}{2},\frac{7}{2},6)$

We can use the GC table to find all these solutions.

2 Inequalities

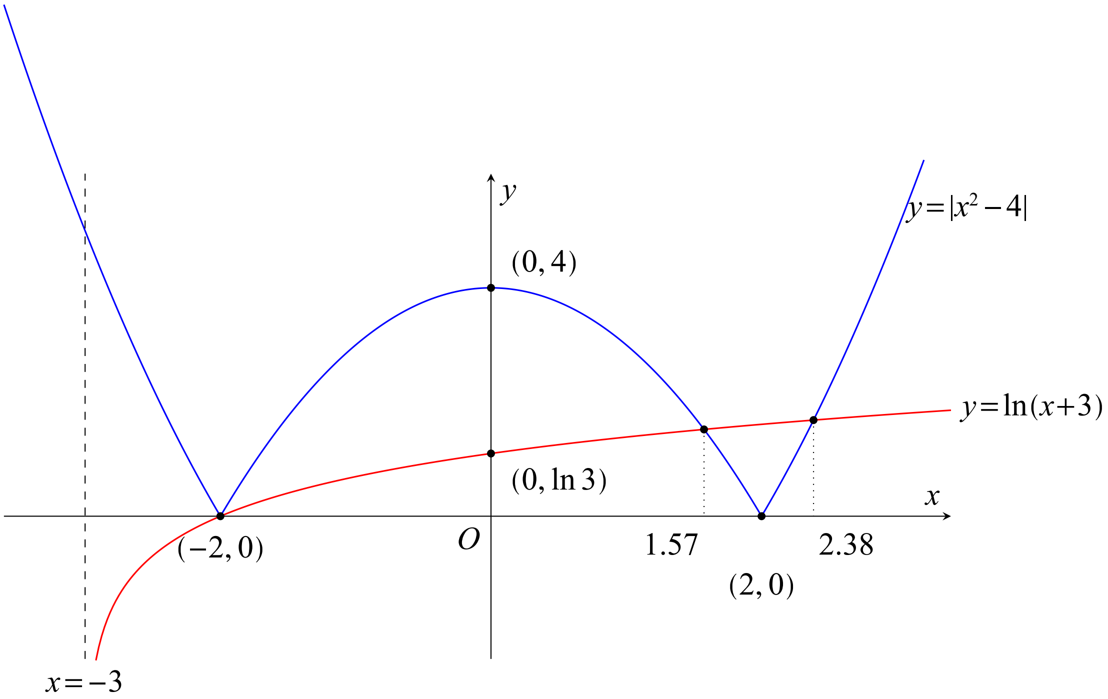
2.1 Solving Inequalities Graphically

Some inequalities cannot be solved algebraically, so we sketch the graphs and read off the solution.

- Rearrange to $\mathrm{f}(x) > 0$, or compare $y = \mathrm{f}(x)$ with $y = \mathrm{g}(x)$.
- Sketch on the GC; mark the intersections and the asymptotes.
- Read off the x-values where the required graph is above or below.

Example 2.1

$$|x^2 - 4| \leqslant \ln(x+3)$$



Using GC, intersections at $x = 1.5746$, $x = 2.3840$

$$\therefore x = -2 \ \text{ or } \ 1.57 \leqslant x \leqslant 2.38$$

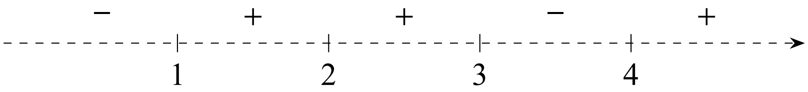
2.2 Sign Test Technique

We solve rational inequalities by finding where the expression changes sign.

- Make the right-hand side 0.
- Do **not** cross multiply unless the expression or bracket is always positive.
- Expressions that are always positive can be divided across, e.g. a quadratic with discriminant <0 and positive leading coefficient: $x^2+2x+5=(x+1)^2+4>0$.

Example 2.2

$$\frac{(x-1)(x-2)^2}{(x-3)(x-4)} \leqslant 0$$



$$x=2:\ \text{numerator } =0\ \ \text{(included)}$$

$$x=3,\ x=4:\ \text{denominator } =0\ \ \text{(excluded)}$$

$$\therefore x\leqslant 1\ \text{ or }\ x=2\ \text{ or }\ 3<x<4$$

You can sketch the graph to check the answer.

> **Note**
>
> i
>
> Crossing a factor with an odd power changes the sign; crossing a factor with an even power keeps the sign the same.

2.3 Common Substitutions

Once an inequality in x is solved, we can replace x with an expression to deduce the solution of a related inequality.

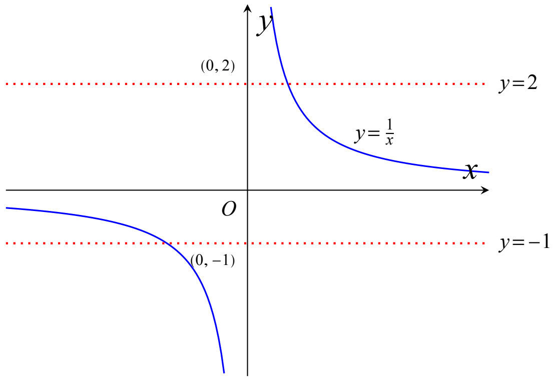
	Original inequality		
Replace x by	$x>2$	$x<-1$	$-1<x<2$
e^x	$x>\ln 2$	no solution	$x<\ln 2$
$\ln x$	$x>e^2$	$0<x<e^{-1}$	$e^{-1}<x<e^2$
$-x$	$x<-2$	$x>1$	$-2<x<1$
$x+1$	$x>1$	$x<-2$	$-2<x<1$
$\lvert x\rvert$	$x<-2$ or $x>2$	no solution	$-2<x<2$
x^2	$x<-\sqrt{2}$ or $x>\sqrt{2}$	no solution	$-\sqrt{2}<x<\sqrt{2}$
$\frac{1}{x}$	$\left(0<x<\frac{1}{2}\right)$	$(-1<x<0)$	$\left(x<-1 \text{ or } x>\frac{1}{2}\right)$
$\cos x$, $0\leqslant x\leqslant 2\pi$	(no solution)	(no solution)	$(0\leqslant x<\pi \text{ or } \pi<x\leqslant 2\pi)$

Regardless of whatever substitution you use, using a graph will always work, but some can be solved without a graph.

For $\frac{1}{x}$, $\sin x$ and $\cos x$, draw the graph to solve.

Example 2.3

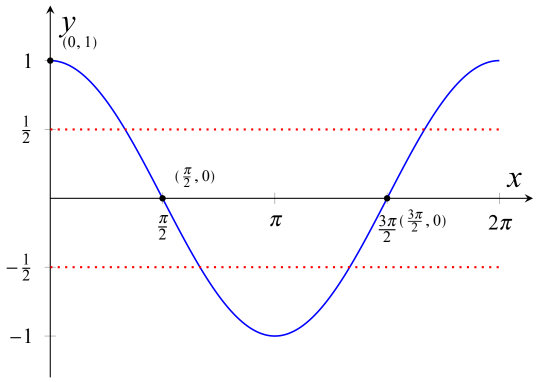
$$\frac{1}{x}>2, \quad \frac{1}{x}<-1, \quad -1<\frac{1}{x}<2$$



Example 2.4

$$\cos x>\tfrac{1}{2}, \quad \cos x<-\tfrac{1}{2},$$

$$-\tfrac{1}{2}<\cos x<\tfrac{1}{2}, \quad 0\leqslant x\leqslant 2\pi$$

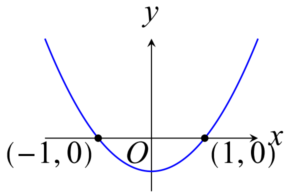


3 Graphing Techniques

3.1 Key Features of a Curve

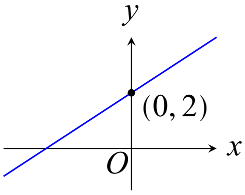
Key features of a curve:

(a) ***x*-intercepts:** let $y=0$.
e.g. $y=x^2-1$: $x^2-1=0 \Rightarrow x=\pm 1$.



***y*-intercepts:** let $x=0$.

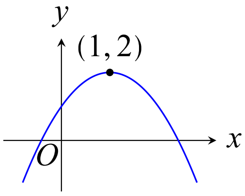
e.g. $y=x+2$: $y=2$.



(b) **Stationary points**, when $\frac{\mathrm{d}y}{\mathrm{d}x}=0$ or $\frac{\mathrm{d}y}{\mathrm{d}x}$ is undefined. Label upon request.

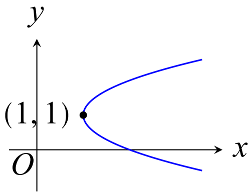
Horizontal turning point: $\frac{\mathrm{d}y}{\mathrm{d}x}=0$ (tangent horizontal).

e.g. $y=-(x-1)^2+2$, maximum point $(1, 2)$.



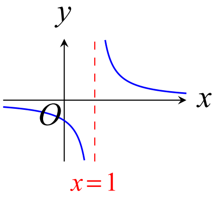
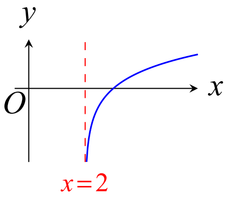
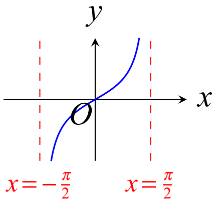
Vertical turning point: $\frac{\mathrm{d}y}{\mathrm{d}x}$ undefined (tangent vertical).

e.g. $x=(y-1)^2+1$, turning point $(1, 1)$.



(c) **Asymptotes**

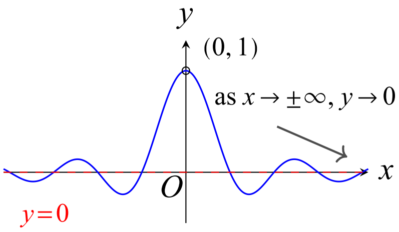
(i) **Vertical asymptotes.** If as $x \to k$, $y \to \pm\infty$, then $x = k$ is a vertical asymptote.

Common Occurrence	**Condition**	**Example and asymptote**	**Graph**
Fractions	denominator $= 0$	$y = \dfrac{1}{x-1}$ $x - 1 = 0 \Rightarrow x = 1$ $\therefore x = 1$ is a vertical asymptote	
Logarithmic graphs	inside $\ln = 0$	$y = \ln(x-2)$ $x - 2 = 0 \Rightarrow x = 2$ $\therefore x = 2$ is a vertical asymptote	
Tangent graphs	inside $\tan = \dfrac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$	$y = \tan x$ $x = \pm\dfrac{\pi}{2},\ \pm\dfrac{3\pi}{2}, \ldots$	

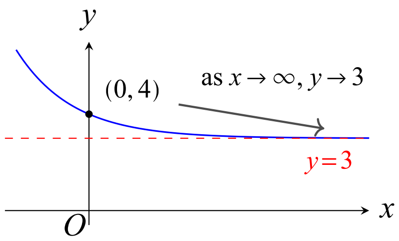
(ii) **Horizontal asymptotes.** As $x \to \pm\infty$, if $y \to k$, then $y = k$ is a horizontal asymptote.

- To find horizontal asymptotes, one should sub in $x \to \pm\infty$ into the equation. This is the only foolproof way to do it.
- A curve **CAN** intersect its horizontal asymptote.

Note: there are more cases of horizontal asymptotes, but these are the common cases.



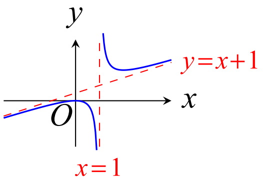
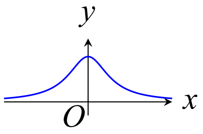
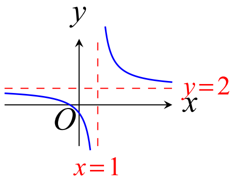
e.g. $y = \dfrac{\sin x}{x}$: curve crosses $y = 0$



e.g. $y = 3 + e^{-x}$

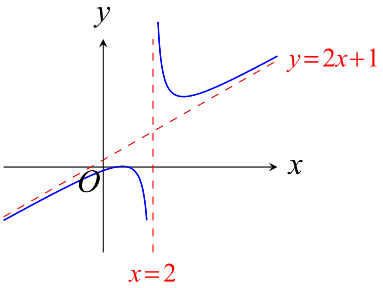
Useful trick to find horizontal asymptotes

For rational functions $\frac{P(x)}{Q(x)}$:

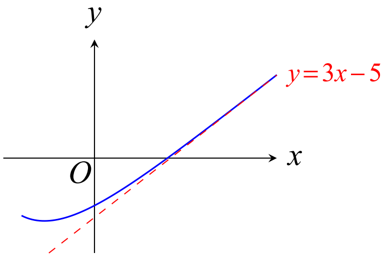
Case	**Horizontal asymptote**	**Example**	**Graph**
numerator power > denominator power	no horizontal asymptote	$y=\frac{x^2}{x-1}$	
numerator power < denominator power	$y=0$	$y=\frac{1}{x^2+1}$	
numerator power = denominator power	ratio of coefficients of highest power	$y=\frac{2x+1}{x-1}$ $y=2$	

(iii) **Oblique asymptotes.** An asymptote of the form $y=mx+c$, $m\neq 0$.

Graphs with oblique asymptotes come in the form $y=mx+c+\mathrm{g}(x)$, where $\mathrm{g}(x)\to 0$ as $x\to\pm\infty$; the oblique asymptote is $y=mx+c$.



e.g. $y=2x+1+\frac{3}{x-2}$



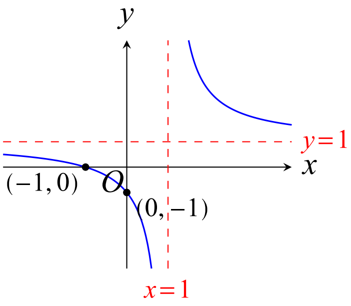
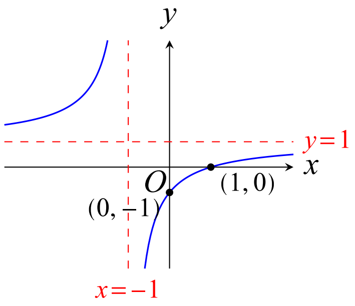
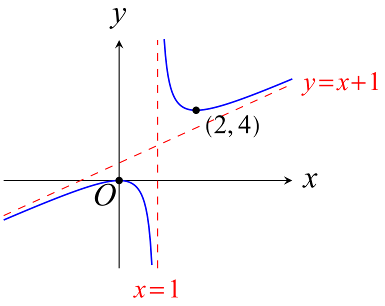
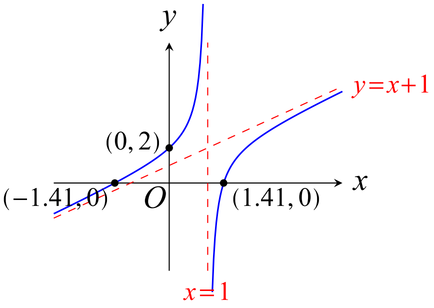
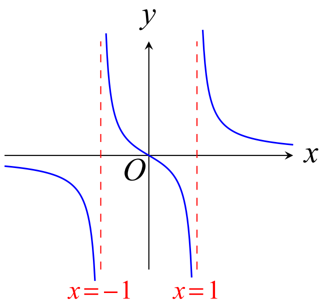
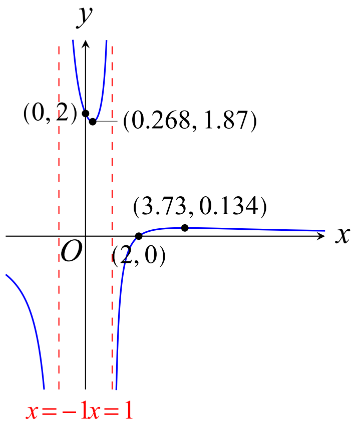
e.g. $y=3x-5+\mathrm{e}^{-x}$

Note

For parametric curves, oblique asymptotes (if they exist) can only be found via the cartesian equation.

3.2 Graphs of Rational Functions

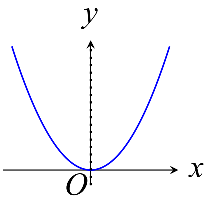
Two common shapes for each type.

Type and form	Shape 1	Shape 2
Linear / linear $y=\frac{Ax+B}{Cx+D}$ Horizontal asymptote: $y=\frac{A}{C}$ Vertical asymptote: $x=-\frac{D}{C}$	 $y=\frac{x+1}{x-1}$	 $y=\frac{x-1}{x+1}$
Quadratic / linear $y=\frac{ax^2+bx+c}{dx+e}$ $=Ax+B+\frac{C}{dx+e}$ Oblique asymptote: $y=Ax+B$ Vertical asymptote: $x=-\frac{e}{d}$	 2 turning points $y=x+1+\frac{1}{x-1}$ $\left(\frac{Cd}{A}>0\right)$	 No turning points $y=x+1-\frac{1}{x-1}$ $\left(\frac{Cd}{A}<0\right)$
Linear / quadratic $y=\frac{ax+b}{cx^2+dx+e}$ Horizontal asymptote: $y=0$ Vertical asymptotes: solve $cx^2+dx+e=0$ (possibly no roots)	 $y=\frac{x}{(x-1)(x+1)}$	 $y=\frac{x-2}{(x-1)(x+1)}$

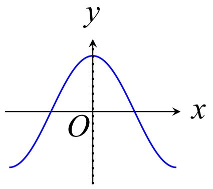
3.3 Symmetry

Even functions: $\mathrm{f}(-x)=\mathrm{f}(x)$.

The graph is symmetric about the y-axis (line of symmetry $x=0$, dotted).



e.g. $y=x^2$

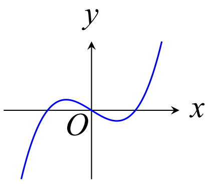


e.g. $y=\cos x$

Odd functions: $\mathrm{f}(-x)=-\mathrm{f}(x)$.

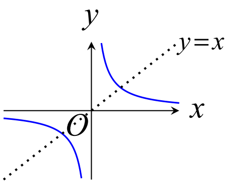
The graph is symmetric about the origin.

e.g. $y=x^3-x$

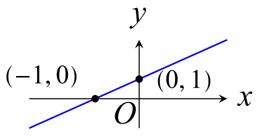
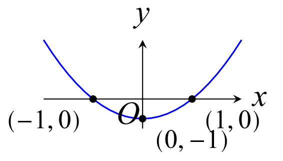
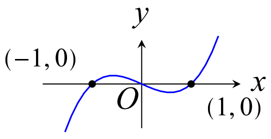
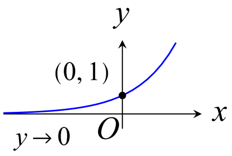
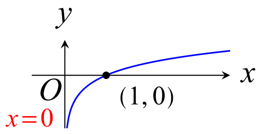
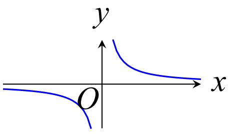
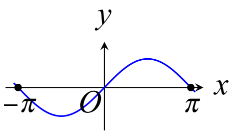
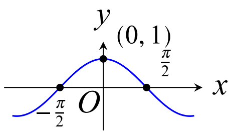
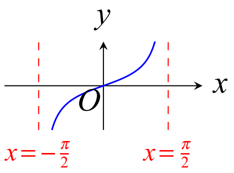
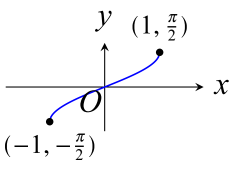
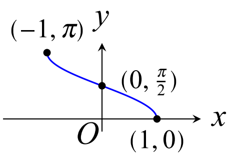
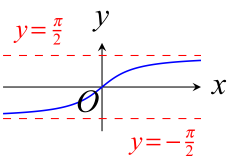


Symmetry about $y=x$: swapping x and y gives the same equation.

e.g. $y=\dfrac{1}{x}$
$(xy=1)$



3.4 Basic Graphs

 Linear $y=x+1$	 Quadratic $y=x^2-1$	 Cubic $y=x^3-x$
 Exponential $y=\mathrm{e}^x$	 Logarithm $y=\ln x$	 Reciprocal $y=\frac{1}{x}$
 $y=\sin x$	 $y=\cos x$	 $y=\tan x$
 $y=\sin^{-1} x$	 $y=\cos^{-1} x$	 $y=\tan^{-1} x$

3.5 Parametric Equations

A curve C can be described by giving both coordinates in terms of a parameter t:

$$x = x(t), \quad y = y(t), \quad t \in T$$

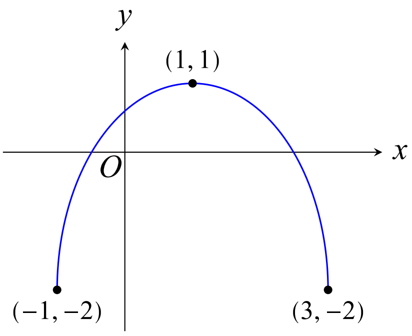
Example 3.1

$x = 2\cos t + 1, \quad y = 3\sin t - 2, \quad 0 \leqslant t \leqslant \pi$

$t = 0$: $(3, -2)$ $\quad t = \frac{\pi}{2}$: $(1, 1)$ $\quad t = \pi$: $(-1, -2)$

$x = 0$: $\left(0, \frac{3\sqrt{3}}{2} - 2\right)$

$y = 0$: $\left(1 \pm \frac{2\sqrt{5}}{3}, 0\right)$



CAUTION: sketching from parametric equations

ALWAYS take note of the range of t when sketching graphs. The restriction on t decides which part of the curve exists: above, $0 \leqslant t \leqslant \pi$ gives only the upper half of the ellipse, with endpoints $(3, -2)$ and $(-1, -2)$, not the whole ellipse.

3.5.1 Finding the Cartesian Equation

The cartesian equation of C is an equation without the parameter t. Common techniques:

① **Make t the subject.** e.g. $x = 2t - 1$, $y = 3t^2 + e^t$, $t \in \mathbb{R}$

$$t = \frac{x+1}{2}$$

$$y = 3\left(\frac{x+1}{2}\right)^2 + e^{\frac{x+1}{2}}$$

② **Trigo identity formulas.** e.g. $x = 2\cos t + 1$, $y = 3\sin t - 2$, $0 \leqslant t \leqslant \pi$

$$\cos t = \frac{x-1}{2}, \quad \sin t = \frac{y+2}{3}$$

$$\cos^2 t + \sin^2 t = 1$$

$$\frac{(x-1)^2}{4} + \frac{(y+2)^2}{9} = 1, \quad y \geqslant -2$$

③ **Others.** e.g. $x = t - \frac{1}{t}$, $y = t + \frac{1}{t}$, $t \neq 0$

$$y + x = 2t, \quad y - x = \frac{2}{t}$$

$$(y+x)(y-x) = 4$$

$$y^2 - x^2 = 4$$

Note: Avoid finding the cartesian eqn if the question did not ask you to.

3.5.2 Asymptotes (Parametric Equations)

- Find (if possible) $t \to t_0$ such that $x \to x_0, y \to \pm\infty$ (vertical asymptote $x = x_0$), or $y \to y_0, x \to \pm\infty$ (horizontal asymptote $y = y_0$).
- Oblique asymptotes (if they exist) can only be found via the cartesian equation.

Example 3.2

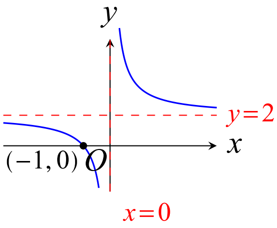
$x = \frac{1}{t-1},\ y = 2t,\ t \neq 1$

As $t \to \pm\infty,\ x \to 0,\ y \to \pm\infty$

$\therefore x = 0$ is a vertical asymptote

As $t \to 1,\ x \to \pm\infty,\ y \to 2$

$\therefore y = 2$ is a horizontal asymptote



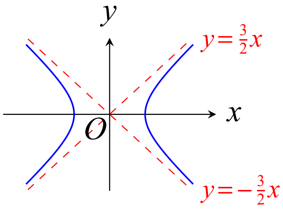
Oblique asymptotes. Cannot be found from the parametric form alone; find the cartesian equation first.

e.g. $x = 2\sec\theta,\ y = 3\tan\theta$

$$\sec^2\theta - \tan^2\theta = 1$$

$$\frac{x^2}{4} - \frac{y^2}{9} = 1$$

Oblique asymptotes: $y = \pm\frac{3}{2}x$



3.5.3 Intersections

Intersection between a parametric curve and another curve: sometimes both equations are needed.

e.g. $x = 2t + \ln t,\ y = t^2 - 1,\ t > 0$ and the circle $x^2 + y^2 = 25$:

$$(2t + \ln t)^2 + (t^2 - 1)^2 = 25$$

Using GC, $t = 0.00735$ or $t = 1.86$ (both > 0)

$$\therefore \text{intersections: } (-4.90,\ -1.00) \text{ and } (4.35,\ 2.47)$$

This equation in t cannot be solved algebraically, so the GC is needed.

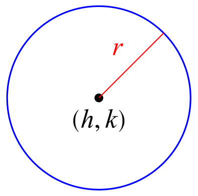
3.6 Conics

Circle

$$(x-h)^2+(y-k)^2=r^2$$

Centre: (h,k)

Radius: r



Parabola

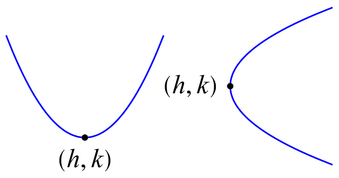
$$y-k=a(x-h)^2 \qquad x-h=b(y-k)^2$$

$y-k=a(x-h)^2$:

turning point (h,k), $\dfrac{\mathrm{d}y}{\mathrm{d}x}=0$

$x-h=b(y-k)^2$:

turning point (h,k), $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ undefined



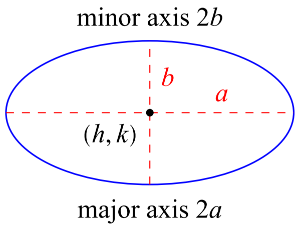
Ellipse

$$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$$

Centre: (h,k)

"Horizontal radius" $=a$, "vertical radius" $=b$

For $a>b$: major axis horizontal (length $2a$, semi-major a); minor axis vertical (length $2b$, semi-minor b).



⋆ RHS ⇒ numerator coefficients of x^2, y^2 must all be 1.

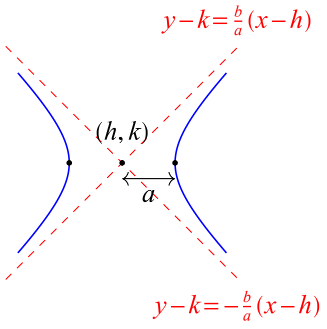
Hyperbola: Type I

$$\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$$

Centre: (h,k)

Distance from centre to vertices $=a$

Asymptotes (dashed): $y-k=\pm\frac{b}{a}(x-h)$


$y-k=\frac{b}{a}(x-h)$
(h,k)
a
$y-k=-\frac{b}{a}(x-h)$

> i
>
> ⋆ RHS ⇒ numerator coefficients of x^2, y^2 must all be 1.
>
> ⋆ Equation of asymptotes: $y-k=\pm\frac{b}{a}(x-h)$

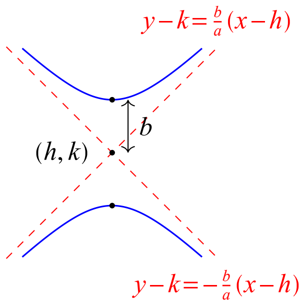
Hyperbola: Type II

$$\frac{(y-k)^2}{b^2}-\frac{(x-h)^2}{a^2}=1$$

Centre: (h,k)

Distance from centre to vertices $=b$

Asymptotes (dashed): $y-k=\pm\frac{b}{a}(x-h)$


$y-k=\frac{b}{a}(x-h)$
b
(h,k)
$y-k=-\frac{b}{a}(x-h)$

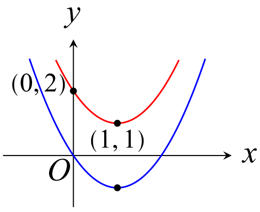
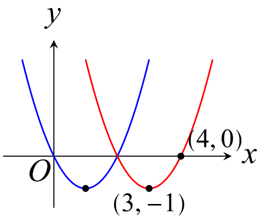
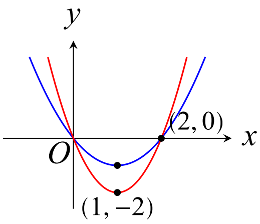
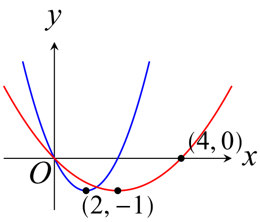
> i
>
> ⋆ RHS ⇒ numerator coefficients of x^2, y^2 must all be 1.
>
> ⋆ Equation of asymptotes: $y-k=\pm\frac{b}{a}(x-h)$

4 Graphing Transformations

Every transformation comes from **replacing** x **or** y in the equation $y=\mathrm{f}(x)$.

Throughout, blue is $y=\mathrm{f}(x)=x(x-2)$, with intercepts $(0,0)$, $(2,0)$ and minimum point $(1,-1)$; red is the transformed graph.

4.1 Translations and Scalings

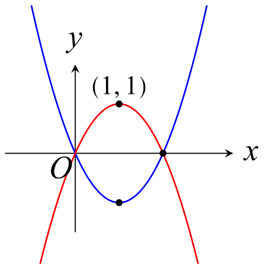
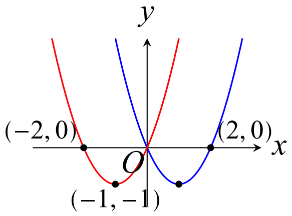
Transformation	**Replacement**	**Result**	**Graph**
Translation of a units in the positive y direction	replace y with $y-a$: $y-a=\mathrm{f}(x)$	$y=\mathrm{f}(x)+a$ (graph: $a=2$)	
Translation of a units in the positive x direction	replace x with $x-a$	$y=\mathrm{f}(x-a)$ (graph: $a=2$)	
Scaling by factor a parallel to the y-axis	replace y with $\frac{y}{a}$: $\frac{y}{a}=\mathrm{f}(x)$	$y=a\mathrm{f}(x)$ (graph: $a=2$)	
Scaling by factor a parallel to the x-axis	replace x with $\frac{x}{a}$	$y=\mathrm{f}\left(\frac{x}{a}\right)$ (graph: $a=2$)	

CAUTION

Every replacement acts the “opposite” way to what it looks like: $x-2$ moves the graph in the **positive** x direction; $y=\mathrm{f}(2x)$ is a scaling by factor $\frac{1}{2}$ (not 2) parallel to the x-axis. Replacing x does **not** mean replacing the input of f, and replacing y does not mean replacing the output $\mathrm{f}(x)$.

If you are ever confused, think of $\sin x$ vs $\sin 2x$.

4.2 Reflections

Transformation	**Replacement**	**Result**	**Graph**
Reflection about the x-axis	replace y with $-y$: $-y=\mathrm{f}(x)$	$y=-\mathrm{f}(x)$	
Reflection about the y-axis	replace x with $-x$	$y=\mathrm{f}(-x)$	

4.3 Combining Transformations

Functions of the form $y=a\mathrm{f}(bx+c)+d$, $a,b,c,d\in\mathbb{R}$: replace x and y **one step at a time**.

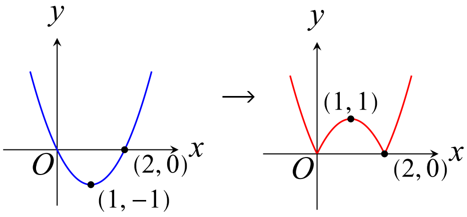
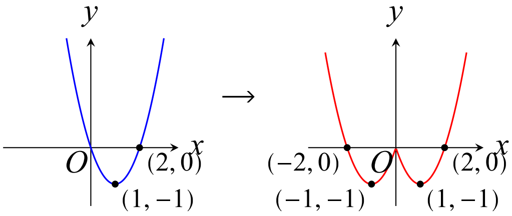
Example 4.1

$y=\mathrm{f}(x) \longrightarrow y=2\mathrm{f}(2x-1)+1$

Replacement	**Transformation**	**Function**
replace x with $x-1$	translation of 1 unit in the positive x direction	$y=\mathrm{f}(x-1)$
replace x with $2x$	scaling by factor $\frac{1}{2}$ parallel to the x-axis	$y=\mathrm{f}(2x-1)$
replace y with $\frac{y}{2}$	scaling by factor 2 parallel to the y-axis	$y=2\mathrm{f}(2x-1)$
replace y with $y-1$	translation of 1 unit in the positive y direction	$y=2\mathrm{f}(2x-1)+1$

4.4 Modulus Graphs

Every modulus graph can be partitioned into a "+" part and a "−" part: $|x|=x$ for $x\geqslant 0$, $|x|=-x$ for $x<0$.

Graph	How to sketch	Graph
$y=\|f(x)\|$	Keep the part on or above the x-axis; reflect the part below the x-axis about the x-axis. Sharp points at the x-intercepts.	
$y=f(\|x\|)$	Remove the part on the left of the y-axis; reflect the part on the right of the y-axis about the y-axis.	

Example 4.2

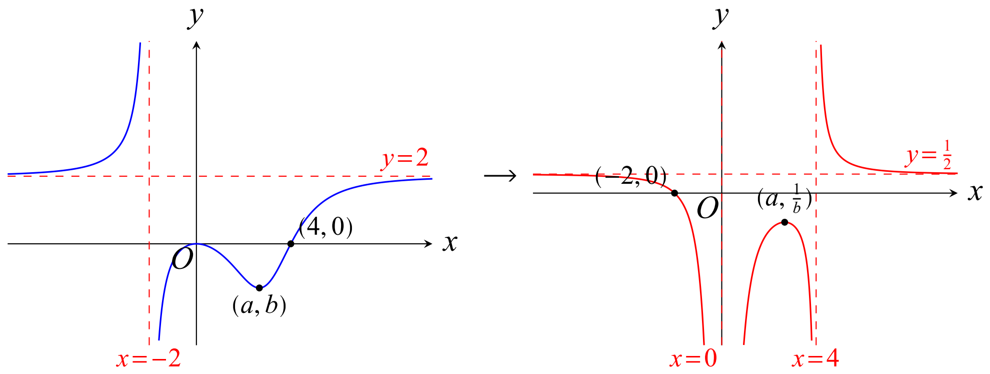
$y=f(x) \longrightarrow y=|f(-2x+1)|$

Replacement	Transformation	Function
replace x with $x+1$	translation of 1 unit in the negative x direction	$y=f(x+1)$
replace x with $-x$	reflection about the y-axis	$y=f(-x+1)$
replace x with $2x$	scaling by factor $\frac{1}{2}$ parallel to the x-axis	$y=f(-2x+1)$
$\|f\|$	reflect parts below the x-axis about the x-axis	$y=\|f(-2x+1)\|$

4.5 Reciprocal Graphs $y=\frac{1}{\mathrm{f}(x)}$

Analyse the new curve at the critical points of $y=\mathrm{f}(x)$.

x-intercept $(a,0)$	$\longleftrightarrow$	vertical asymptote $x=a$
vertical asymptote $x=a$	$\longleftrightarrow$	$y\to 0$: point $(a,0)$ (excluded)
maximum point (a,b), $b\neq 0$	$\longleftrightarrow$	minimum point $\left(a,\frac{1}{b}\right)$
minimum point (a,b), $b\neq 0$	$\longleftrightarrow$	maximum point $\left(a,\frac{1}{b}\right)$
horizontal asymptote $y=k$, $k\neq 0$	$\longleftrightarrow$	horizontal asymptote $y=\frac{1}{k}$
$y\to\pm\infty$ (incl. oblique asymptote)	$\longleftrightarrow$	horizontal asymptote $y=0$
$y=\pm 1$	$\longleftrightarrow$	same point $(\frac{1}{\pm 1}=\pm 1)$
$\mathrm{f}(x)>0$ / $\mathrm{f}(x)<0$	$\longleftrightarrow$	$\frac{1}{\mathrm{f}(x)}>0$ / $\frac{1}{\mathrm{f}(x)}<0$ (same sign)



$y=\mathrm{f}(x)$

$y=\dfrac{1}{\mathrm{f}(x)}$

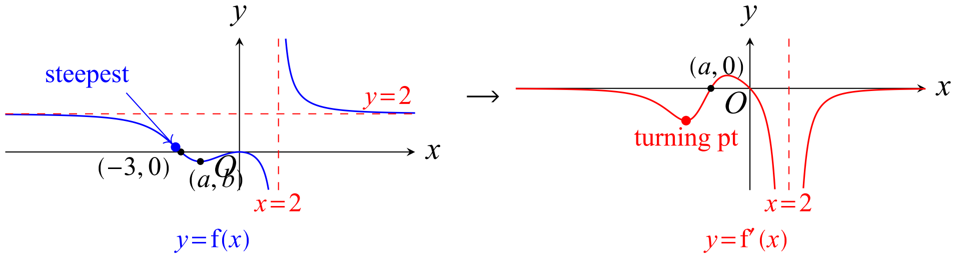
Note: sketching $y=\frac{1}{\mathrm{f}(x)}$

- Same sign: $\mathrm{f}>0\Rightarrow\frac{1}{\mathrm{f}}>0$, $\mathrm{f}<0\Rightarrow\frac{1}{\mathrm{f}}<0$.
- $|\mathrm{f}|$ large $\Rightarrow\left|\frac{1}{\mathrm{f}}\right|$ small; $|\mathrm{f}|$ small $\Rightarrow\left|\frac{1}{\mathrm{f}}\right|$ large.
- $\mathrm{f}=0\Rightarrow\frac{1}{\mathrm{f}}$ undefined (vertical asymptote).
- $\mathrm{f}=\pm 1\Rightarrow\frac{1}{\mathrm{f}}=\pm 1$ (graphs meet).

4.6 Derivative Graphs $y = f'(x)$

$f'(x) = \dfrac{dy}{dx}$ is the gradient of $y = f(x)$: read off the gradient on each interval.

$y = f(x)$	$y = f'(x)$
stationary point at $x = a$	x-intercept $(a, 0)$
increasing / decreasing	above / below the x-axis
point of inflexion (max. or min. gradient)	turning point
vertical asymptote $x = a$	vertical asymptote $x = a$
horizontal asymptote $y = k$	horizontal asymptote $y = 0$
oblique asymptote $y = mx + c$	horizontal asymptote $y = m$



The turning point on the derivative graph corresponds to the steepest point on the original graph.

CAUTION

Mark x-coordinates only where they come from $y = f(x)$: the y-values of $y = f'(x)$ are generally **not** known from the graph of $y = f(x)$.

4.7 Reverse Transformations

If $y = f(x)$ undergoes a sequence of transformations to give a known $y = g(x)$, undo the transformations **in reverse order**, each with its opposite replacement.

Example 4.3

$y = f(x)$ undergoes, in order, the following to give $y = 6e^{-x+3}$. Find $f(x)$.

A: translation of 2 units in the negative x direction

B: reflection about the y-axis

C: scaling by factor 3 parallel to the y-axis

Step	Opposite transformation	Replacement	Function
C^{-1}	scale by factor $\frac{1}{3}$ parallel to the y-axis	replace y with $3y$	$y = 2e^{-x+3}$
B^{-1}	reflect about the y-axis	replace x with $-x$	$y = 2e^{x+3}$
A^{-1}	translate 2 units in the positive x direction	replace x with $x-2$	$y = 2e^{x+1}$

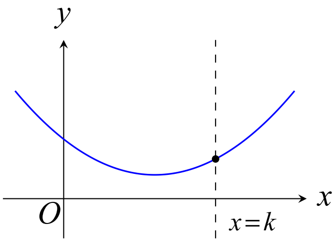
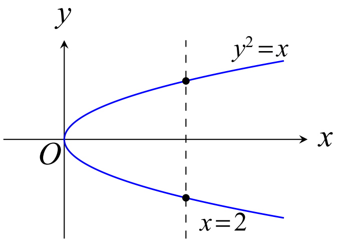
$$\therefore f(x) = 2e^{x+1}$$

5 Functions

5.1 Functions, Domain and Range

$$\mathrm{f}: x \mapsto \underbrace{x^2+3x+2}_{\text{rule } \mathrm{f}(x)}, \quad x \in \mathbb{R}, \quad \underbrace{x \geqslant 0}_{\text{domain } D_\mathrm{f}}$$

A function is a relation that gives **exactly one** value of y for each $x \in D_\mathrm{f}$.

It's a function	**Not a function**
 every vertical line $x=k$, $k \in D_\mathrm{f}$, cuts the graph **exactly once**	 one vertical line ($x=2$) cuts the graph **more than once**

Inequality	**Set notation**	**Interval**
$a<x<b$	$\{x \in \mathbb{R} : a<x<b\}$	(a, b)
$a \leqslant x \leqslant b$	$\{x \in \mathbb{R} : a \leqslant x \leqslant b\}$	$[a, b]$
$a \leqslant x < b$	$\{x \in \mathbb{R} : a \leqslant x < b\}$	$[a, b)$
$x>a$	$\{x \in \mathbb{R} : x>a\}$	(a, ∞)
$x \in \mathbb{R},\ x \neq 0$	$\mathbb{R} \setminus \{0\}$	$(-\infty, 0) \cup (0, \infty)$

CAUTION

To find the range of a function, **ALWAYS** sketch its graph $y=\mathrm{f}(x)$, $x \in D_\mathrm{f}$, and read the range off the graph. **Never** just sub in the endpoints of the domain.

5.2 Inverse Functions

	Working
Exists	**Exists:** f is one-to-one: every line $y=k$, $k\in\mathbb{R}$, cuts $y=\mathrm{f}(x)$ **at most once** $\Rightarrow$ f^{-1} exists. $\Rightarrow$ f is one-to-one, thus f^{-1} exists. **Does not exist:** e.g. $\mathrm{f}(x)=(x-3)^2-7$, $x\in\mathbb{R}$: the line $y=0$ cuts $y=\mathrm{f}(x)$ **twice** (at $x=3\pm\sqrt{7}$) $\Rightarrow$ f^{-1} does not exist. Alternatively: $\exists\, x_1\neq x_2$ such that $\mathrm{f}(x_1)=\mathrm{f}(x_2)$ $\therefore$ f is not one-one $\Rightarrow$ f^{-1} does not exist.
Rule	make x the subject of $y=\mathrm{f}(x)$
Domain	$D_{\mathrm{f}^{-1}}=R_\mathrm{f}$
Range	$R_{\mathrm{f}^{-1}}=D_\mathrm{f}$
Graph	reflection of $y=\mathrm{f}(x)$ about $y=x$: all x- and y-coordinates are swapped, $(a,b)\rightarrow(b,a)$

Example 5.1

$\mathrm{f}:x\mapsto(x-3)^2-7,\ x\geqslant 3$

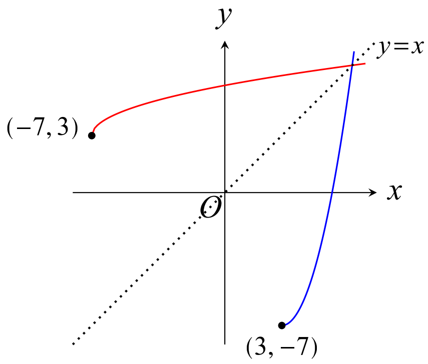
$$y=(x-3)^2-7$$

$$x-3=\sqrt{y+7}\quad(x\geqslant 3)$$

$$x=3+\sqrt{y+7}$$

$$R_\mathrm{f}=[-7,\infty)$$

$$\therefore\ \mathrm{f}^{-1}:x\mapsto 3+\sqrt{x+7},\ x\in\mathbb{R},\ x\geqslant -7$$



CAUTION

- When drawing $y=\mathrm{f}(x)$ and $y=\mathrm{f}^{-1}(x)$ together, use the **same scale** on both axes.
- $\mathrm{ff}^{-1}(x)=\mathrm{f}^{-1}\mathrm{f}(x)=x$, but $D_{\mathrm{ff}^{-1}}=D_{\mathrm{f}^{-1}}=R_\mathrm{f}$ while $D_{\mathrm{f}^{-1}\mathrm{f}}=D_\mathrm{f}$.
- Solving $\mathrm{f}(x)=\mathrm{f}^{-1}(x)$: it is **not always** $\mathrm{f}(x)=x$ (only when the curves meet on $y=x$).

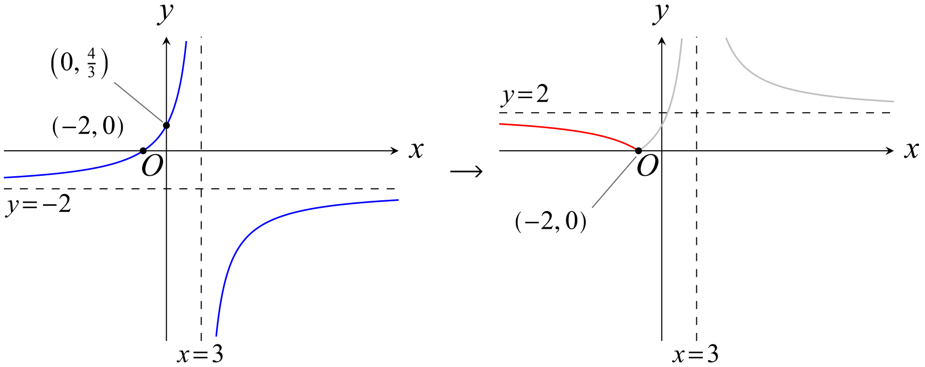
5.2.1 Removing the Modulus

On a restricted domain, the sign inside the modulus is fixed, so the modulus can be removed: $|x-a|=x-a$ for $x\geqslant a$, $|x-a|=-(x-a)$ for $x<a$.

Example 5.2

$\mathrm{f}:x\to\left|\dfrac{2x+4}{3-x}\right|$, $x\in\mathbb{R}$, $x\leqslant-2$ (2023 A-Level P1 Q7, with the domain restricted to $x\leqslant a$, $a=-2$)

$$\begin{aligned} y=\mathrm{f}(x)&=\left|\frac{2x+4}{3-x}\right| \\ &=-\frac{2x+4}{3-x} \qquad (x\leqslant-2) \end{aligned}$$



$y=\dfrac{2x+4}{3-x}$

$y=\left|\dfrac{2x+4}{3-x}\right|$, $x\leqslant-2$

Note

Whenever you are asked to find f^{-1} and there is a modulus, you are usually able to remove it in the first step by determining whether the region of the graph is positive or negative for the domain of f.

5.3 Composite Functions

	Working
Exists	**Exists:** $R_{\mathrm{g}} \subseteq D_{\mathrm{f}}$ **Does not exist:** $x_0 \in R_{\mathrm{g}},\ x_0 \notin D_{\mathrm{f}}\ \therefore R_{\mathrm{g}} \not\subseteq D_{\mathrm{f}}$
Rule	sub $\mathrm{g}(x)$ into $\mathrm{f}(x)$: $\mathrm{f}[\mathrm{g}(x)]$
Domain	$D_{\mathrm{fg}} = D_{\mathrm{g}}$: rightmost function's domain (e.g. $D_{\mathrm{fgh}} = D_{\mathrm{h}}$)
Range	$D_{\mathrm{g}} \xrightarrow{\mathrm{g}} R_{\mathrm{g}} \xrightarrow{\mathrm{f}} R_{\mathrm{fg}}$: sketch $y = \mathrm{f}(x)$ on the domain R_{g} Alternatively, sketch the graph of $\mathrm{fg}(x)$ for D_{fg} (avoid, as fg might usually be quite hard to deal with).

5.3.1 Composite with a Piecewise Function

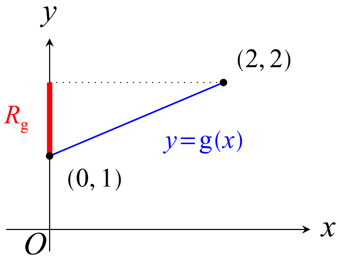
Split according to which piece of f the value $\mathrm{g}(x)$ lands in.

Example 5.3

$$\mathrm{f}(x) = \begin{cases} x^2, & 0 \leqslant x < 1 \\ 2 - x, & 1 \leqslant x \leqslant 2 \end{cases} \qquad \mathrm{g}(x) = \frac{x}{2} + 1,\ 0 \leqslant x \leqslant 2$$

$$R_{\mathrm{g}} = [1, 2]$$

$$\therefore\ \mathrm{fg}(x) = \mathrm{f}(\mathrm{g}(x)) = 2 - \left(\frac{x}{2} + 1\right) = 1 - \frac{x}{2}, \quad 0 \leqslant x \leqslant 2$$

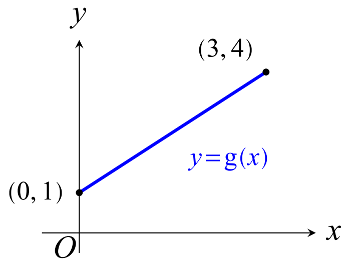


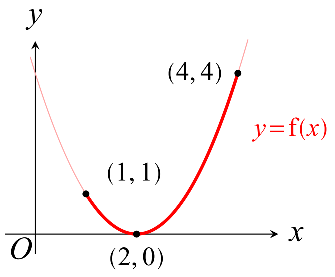
Example 5.4

$\mathrm{g}(x) = x + 1,\ 0 \leqslant x \leqslant 3$; $\quad \mathrm{f}(x) = (x-2)^2,\ x \in \mathbb{R}$. Assume fg exists.

$$\begin{array}{ccccc} D_{\mathrm{g}} & \xrightarrow{\mathrm{g}} & R_{\mathrm{g}} & \xrightarrow{\mathrm{f}} & R_{\mathrm{fg}} \\ [0, 3] & & [1, 4] & & [0, 4] \end{array}$$

$$\therefore\ R_{\mathrm{fg}} = [0, 4]$$





5.4 Functional Powers and Self-Inverse Functions

$f^2(x) = f[f(x)]$ exists if $R_f \subseteq D_f$; similarly f^3, f^4, … Note $f^2(x) \neq [f(x)]^2$.

Self-inverse: $f = f^{-1} \iff f^2(x) = x$ for all $x \in D_f$ (graph symmetric about $y = x$).

Periodic composition pattern. If $g^2(x) = x$:

$$g^n(x) = \begin{cases} g(x), & n \text{ odd} \\ x, & n \text{ even} \end{cases}$$

$$g^{2027}(x) = g(x).$$

Similarly, if $f^3(x) = x$, the pattern repeats every 3: $f^n = f^{n \bmod 3}$, and $f^{-1} = f^2$.

5.5 Piecewise and Periodic Functions

$$f : x \mapsto \begin{cases} f_1(x), & x \in D_1 \\ f_2(x), & x \in D_2 \end{cases}, \qquad f(x) = f(x+p)$$

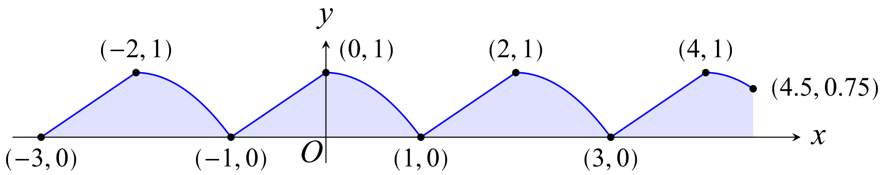
$f(x) = f(x+p)$: the graph repeats every p units (period p).

→ Always sketch the **original copy** first, then copy & paste.

→ Never sub any x value into f that is outside the domain of the original copy: shift by multiples of p first.

Example 5.5

$$f(x) = \begin{cases} 1 - x^2, & 0 \leqslant x < 1 \\ x - 1, & 1 \leqslant x < 2 \end{cases}, \quad f(x) = f(x+2), \quad -3 \leqslant x \leqslant 4.5$$



$$f(4.5) = f(4.5 - 4) = f(0.5) = 1 - 0.5^2 = 0.75$$

Area under a periodic function: area over n full periods $= n \times$ area over one period.

$$\begin{aligned} \int_{-3}^{4.5} f(x)\,dx &= \int_{-3}^{3} f(x)\,dx + \int_{3}^{4} f(x)\,dx + \int_{4}^{4.5} f(x)\,dx \\ &= 3\int_{0}^{2} f(x)\,dx + \int_{1}^{2} (x-1)\,dx + \int_{0}^{0.5} (1 - x^2)\,dx \\ &= \frac{107}{24} \approx 4.46 \end{aligned}$$

5.5.1 Inverse of a Piecewise Function

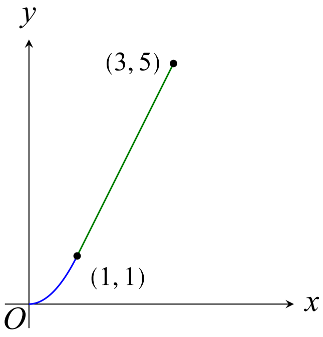
Check one-to-one using the ranges of the pieces (they must not overlap and each piece is one-to-one); then invert each piece, with domain = range of that piece.

Example 5.6

$$\mathrm{h}(x)=\begin{cases} x^2, & 0\leqslant x\leqslant 1 \\ 2x-1, & 1<x\leqslant 3 \end{cases}$$

ranges: $[0,1]$ and $(1,5]$ (no overlap)

$$\mathrm{h}^{-1}(x)=\begin{cases} \sqrt{x}, & 0\leqslant x\leqslant 1 \\ \dfrac{x+1}{2}, & 1<x\leqslant 5 \end{cases}$$



6 Sequences and Series

A **sequence** is a set of numbers arranged in a definite order, A **series** is the sum of the terms of a sequence:

$$u_1,\ u_2,\ u_3,\ \ldots,\qquad u_n\text{: the } n\text{th term}$$

$$S_n = u_1 + u_2 + \cdots + u_n = \sum_{r=1}^{n} u_r, \qquad S_\infty = u_1 + u_2 + u_3 + \cdots = \sum_{r=1}^{\infty} u_r \quad \text{(infinite series)}$$

6.1 Recurrence Relations

A recurrence relation defines u_n in terms of one or more preceding terms, together with the initial condition(s). **First-order recurrence relation**: requires only one initial condition; defined using only the immediately preceding term, i.e. u_{n-1} only.

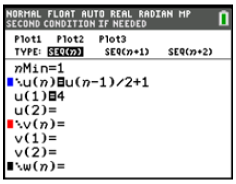
Second-order recurrence relation: requires two initial conditions; defined using the two preceding terms, i.e. u_{n-1}, u_{n-2}.

Example 6.1

$$u_{r+1} = \frac{u_r}{2} + 1, \qquad u_1 = 4$$

$$u_2 = 3, \quad u_3 = \frac{5}{2}, \quad u_4 = \frac{9}{4}$$

Rule: $u_{r+1} = \frac{u_r}{2} + 1$; initial condition: $u_1 = 4$.



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NORMAL FLOAT AUTO REAL RADIAN MP
SECOND CONDITION IF NEEDED
 Plot1  Plot2  Plot3
 TYPE: SEQ(n)  SEQ(n+1)  SEQ(n+2)
 nMin=1
■ u(n)=u(n−1)/2+1
 u(1)=4
 u(2)=
■ v(n)=
 v(1)=
 v(2)=
■ w(n)=
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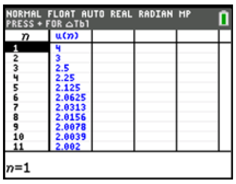

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n	u(n)
1	4
2	3
3	2.5
4	2.25
5	2.125
6	2.0625
7	2.0313
8	2.0156
9	2.0078
10	2.0039
11	2.002

n=1

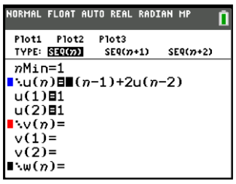
Example 6.2

$$u_{r+1} = 2u_r + 1, \qquad u_1 = 1$$

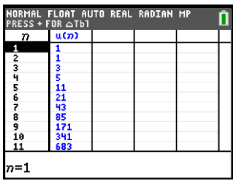
Example 6.3

$$u_{n+2} = u_{n+1} + 2u_n, \qquad u_1 = 1, \qquad u_2 = 1$$

$$1,\ 1,\ 3,\ 5,\ 11,\ 21,\ 43,\ \ldots$$



```
NORMAL FLOAT AUTO REAL RADIAN MP
 Plot1  Plot2  Plot3
 TYPE: SEQ(n)  SEQ(n+1)  SEQ(n+2)
 nMin=1
■ u(n)=■(n−1)+2u(n−2)
 u(1)=1
 u(2)=1
■ v(n)=
 v(1)=
 v(2)=
■ w(n)=
```



NORMAL FLOAT AUTO REAL RADIAN MP
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n	u(n)
1	1
2	1
3	3
4	5
5	11
6	21
7	43
8	85
9	171
10	341
11	683

n=1

Example 6.4

(GP)

$$u_{n+1} = ru_n, \qquad u_1 = a$$

$$a,\ ar,\ ar^2,\ \ldots$$

Example 6.5

(AP)

$$u_{n+1} = u_n + d, \qquad u_1 = a$$

$$a,\ a+d,\ a+2d,\ \ldots$$

(However, this is not unique.)

$$u_{n+2} - 2u_{n+1} + u_n = 0, \qquad u_1 = a, \qquad u_2 = a+d$$

6.1.1 Limit of a Sequence

$$\text{As } r \to \infty,\ u_r = \frac{1}{r} \to 0 \quad \text{or} \quad \lim_{r\to\infty}(u_r) = \lim_{r\to\infty}\left(\frac{1}{r}\right) = 0$$

$$\text{As } r \to \infty,\ u_r \to L \quad \text{or} \quad \lim_{r\to\infty}(u_r) = L \quad \text{(convergent)}$$

$u_r \to L \Rightarrow u_{r+1} \to L,\ u_{r+2} \to L,\ \ldots$

(For recurrence relations: if the sequence is given to be convergent, change all u_n, u_{n+1}, u_{n+2} to L and solve for L.)

Rules (lim u_r, lim v_r real):

$$\lim_{r\to\infty}(a) = a$$

$$\lim_{r\to\infty}(u_r \pm v_r) = \lim_{r\to\infty}(u_r) \pm \lim_{r\to\infty}(v_r)$$

$$\lim_{r\to\infty}(u_r v_r) = \lim_{r\to\infty}(u_r) \times \lim_{r\to\infty}(v_r)$$

$$\lim_{r\to\infty}\left(\frac{u_r}{v_r}\right) = \frac{\lim_{r\to\infty}(u_r)}{\lim_{r\to\infty}(v_r)}, \quad \lim_{r\to\infty}(v_r) \neq 0$$

6.2 Sigma Notation

$$\sum_{r=m}^{n} \mathrm{f}(r) = \mathrm{f}(m) + \mathrm{f}(m+1) + \cdots + \mathrm{f}(n)$$

$$S_n = \sum_{r=1}^{n} u_r = u_1 + u_2 + \cdots + u_n$$ (By default, S_n starts from $r=1$ unless otherwise stated.)

Properties:

$$\text{no. of terms in } \sum_{r=m}^{n} u_r :\ n-m+1$$

$$\sum_{r=m}^{n} a = (n-m+1)a$$

$$\sum_{r=m}^{n} a u_r = a \sum_{r=m}^{n} u_r$$

$$\sum_{r=m}^{n} (u_r \pm v_r) = \sum_{r=m}^{n} u_r \pm \sum_{r=m}^{n} v_r$$

$$\sum_{r=m}^{n} u_r = \sum_{r=1}^{n} u_r - \sum_{r=1}^{m-1} u_r$$

(To remember the last three, note that the properties are similar to what you can do in integration.)

6.3 Standard Formulas

$$\boxed{\sum_{r=1}^{n} r = \frac{n(n+1)}{2}}$$

AP with first term 1, common difference 1:

$$\sum_{r=1}^{n} r = 1+2+\cdots+n$$

$$S_n = \frac{n}{2}\left[2(1)+(n-1)(1)\right]$$

$$= \frac{n(n+1)}{2}$$

$$\boxed{\sum_{r=1}^{n} a^r = \frac{a\left(a^n-1\right)}{a-1}, \quad a \neq 1}$$

GP with first term a, common ratio a:

$$\sum_{r=1}^{n} a^r = a + a^2 + \cdots + a^n$$

$$= \frac{a\left(a^n-1\right)}{a-1}$$

$$\boxed{\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}}$$

$$\boxed{\sum_{r=1}^{n} r^3 = \left[\frac{n(n+1)}{2}\right]^2 = \frac{n^2(n+1)^2}{4}}$$

> **Note**
>
> $\sum r^2$ and $\sum r^3$ are given in the exam if needed. All four results hold only if r starts from 1.

6.4 Augmented Forms

Example 6.6

$$\sum_{r=1}^{n} \frac{4}{5^{2r+1}} = 4\sum_{r=1}^{n} \frac{1}{5 \cdot 25^r}$$

$$= \frac{4}{5}\sum_{r=1}^{n}\left(\frac{1}{25}\right)^r$$

$$= \frac{4}{5} \cdot \frac{\frac{1}{25}\left[1-\left(\frac{1}{25}\right)^n\right]}{1-\frac{1}{25}}$$

$$= \frac{4}{5} \cdot \frac{1}{24}\left[1-\left(\frac{1}{25}\right)^n\right]$$

$$= \frac{1}{30}\left(1-\frac{1}{25^n}\right)$$

> **Note**
>
> To use the $\sum a^r$ formula, the power must be exactly r, not $r+1$, not $2r$. Rewrite first.

6.5 Substitution

$$\sum_{r=m}^{n} \mathrm{f}(r) = \sum_{r=m-k}^{n-k} \mathrm{f}(r+k) \quad \text{(replace } r \text{ with } r+k\text{)}$$

Example 6.7

Given $\displaystyle\sum_{r=2}^{n} \frac{1}{r(r+2)} = \frac{1}{2}\left(\frac{5}{6} - \frac{1}{n+1} - \frac{1}{n+2}\right)$, find $\displaystyle\sum_{r=2}^{n} \frac{1}{r^2-1}$.

$$\begin{aligned}
\sum_{r=2}^{n} \frac{1}{r^2-1} &= \sum_{r=2}^{n} \frac{1}{(r-1)(r+1)} \\
&= \sum_{r=1}^{n-1} \frac{1}{r(r+2)} \quad \text{(replace } r \text{ with } r+1\text{)} \\
&= \sum_{r=1}^{1} \frac{1}{r(r+2)} + \sum_{r=2}^{n-1} \frac{1}{r(r+2)} \\
&= \frac{1}{3} + \frac{1}{2}\left(\frac{5}{6} - \frac{1}{n} - \frac{1}{n+1}\right) \\
&= \frac{3}{4} - \frac{1}{2n} - \frac{1}{2(n+1)}
\end{aligned}$$

Note

Substitution only allows **translation** of r values, e.g. replace r with …, $r-2$, $r-1$, $r+1$, $r+2$, …

Scaling is not allowed. E.g. replacing r with $2r$ is wrong. List out the terms to see why.

6.6 Convergence

Example 6.8

Given $u_n=\dfrac{1}{n(n+1)}$ and $S_n=\sum_{r=1}^{n} u_r=1-\dfrac{1}{n+1}$.

Convergence of u_n**:** let $n\to\infty$ in u_n.	**Convergence of** S_n**:** let $n\to\infty$ in S_n.
$u_n=\dfrac{1}{n(n+1)}$ As $n\to\infty,\ u_n\to 0$	$S_n=\sum_{r=1}^{n}\dfrac{1}{r(r+1)}$ $=1-\dfrac{1}{n+1}$ As $n\to\infty,\ S_n\to 1$ (finite) $\therefore$ the series converges, $S_\infty=1$

Note

(1) $u_n\to 0$ is not strong enough evidence that S_n converges, e.g. $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots=\infty$.

(2) However, if S_n converges, then u_n **must** converge to 0.

6.7 Finding u_n or S_n from Each Other

6.7.1 Finding u_n from S_n

$$u_n=S_n-S_{n-1},\quad u_1=S_1$$

Example 6.9

$$\begin{aligned}S_n&=n^2+2n\\ u_n&=S_n-S_{n-1}\\ &=(n^2+2n)-[(n-1)^2+2(n-1)]\\ &=2n+1\end{aligned}$$

6.7.2 Finding S_n from u_n

Example 6.10

$$\begin{aligned}u_r&=3r^2+r\\ S_n&=\sum_{r=1}^{n}(3r^2+r)\\ &=3\sum_{r=1}^{n}r^2+\sum_{r=1}^{n}r\\ &=3\cdot\frac{n(n+1)(2n+1)}{6}+\frac{n(n+1)}{2}\end{aligned}$$

7 Arithmetic and Geometric Progressions

An **arithmetic progression (AP)** is a sequence with a common difference d, e.g. $2,\ 5,\ 8,\ 11,\ \ldots$

A **geometric progression (GP)** is a sequence with a common ratio r, e.g. $3,\ 6,\ 12,\ 24,\ \ldots$

u_n: the nth term, S_n: the sum of the first term up to the nth term.

	AP	**GP**
Property	common difference d	common ratio r
Example	$2,\ 5,\ 8,\ 11,\ \ldots \quad (d=3)$	$3,\ 6,\ 12,\ 24,\ \ldots \quad (r=2)$
u_n	$u_n=a+(n-1)d$	$u_n=ar^{n-1}$
S_n	$S_n=\frac{n}{2}\,[2a+(n-1)d]$ $S_n=\frac{n}{2}\,(a+l)$	$S_n=\frac{a\,(1-r^n)}{1-r}=\frac{a\,(r^n-1)}{r-1},\ r\neq 1$
S_∞	not convergent	$S_\infty=\frac{a}{1-r}$, only if $\lvert r\rvert<1$
Proving	$u_{n+1}-u_n=d$ (constant) $u_3-u_2=k_1,\ u_2-u_1=k_2$ $k_1\neq k_2\ \therefore$ not an AP	$\frac{u_{n+1}}{u_n}=r$ (constant) $\frac{u_3}{u_2}=k_1,\ \frac{u_2}{u_1}=k_2$ $k_1\neq k_2\ \therefore$ not a GP

However, to prove that something IS an AP, you have to do it in terms of n. You cannot compare u_1, u_2 and u_3 in the same way that we prove that something is not an AP.

Note: AP: $u_2,\ u_4,\ u_6,\ \ldots$ is an AP (common difference $2d$), GP: $u_2,\ u_4,\ u_6,\ \ldots$ is a GP (common ratio r^2).

Important Note

Remember that you can use a GC table whenever needed, especially when the variable you are solving for is an integer. It is very common in this chapter because a lot of the time you are solving for $n\in\mathbb{Z}^+$.

7.1 Summing

Example 7.1

AP: first term 9, common difference 2. Sum of the 20th to 50th terms inclusive:

$$\begin{aligned}S_{50}-S_{19}&=\frac{50}{2}\,[2(9)+49(2)]-\frac{19}{2}\,[2(9)+18(2)]\\&=2900-513\\&=2387\end{aligned}$$

7.2 Common Questions

7.2.1 Comparing AP and GP terms

Example 7.2

[TMJC 2025 Prelim P1 Q4]

An arithmetic series has first term a and common difference d, where a and d are non-zero real constants. A convergent geometric series has first term b and common ratio r, where $b>0$ and r is a non-zero real constant. The sum of the first n terms of the geometric series is denoted by S.

It is given that the 7th, 10th and 11th terms of the arithmetic series are equal to the 3rd, 6th and 10th terms of the geometric series respectively.

$$a+6d=br^2 \quad -- \ (1)$$

$$a+9d=br^5 \quad -- \ (2)$$

$$a+10d=br^9 \quad -- \ (3)$$

7.2.2 Bank Questions

Important Note

i Any bank questions will come in the form $u_{n+1}=pu_n+q$, where p and q are constants.

Example 7.3

[TJC 2025 Prelim P1 Q7]

Janice, on the other hand, deposits \$100 at the start of every month into a bank account that earns 0.3% interest per month, with interest calculated and added into the account at the end of each month.

Find the amount that Janice will have in her savings account at the end of N months.

nth mth	Amount at start of month (\$)	Amount at end of month (\$)
1 Jan26	100	$100(1.003)$
2	$100+100(1.003)$ $=100(1+1.003)$	$100(1+1.003)(1.003)$ $=100\,(1.003+1.003^2)$
3	$100+100\,(1.003+1.003^2)$ $=100\,(1+1.003+1.003^2)$	$100\,(1+1.003+1.003^2)\,(1.003)$ $=100\,(1.003+1.003^2+1.003^3)$
$\vdots$	$\vdots$	$\vdots$
N		$100\,(1.003+1.003^2+\cdots+1.003^N)$ $=\dfrac{100(1.003)\,(1.003^N-1)}{1.003-1}$

7.2.3 Contextual Questions

For general contextual questions, it is important to determine whether the question is asking for u_n or S_n.

Example 7.4

[RVHS 2025 Prelim P2 Q4]

The dosage of medicine given to a patient needs to be carefully managed in order to achieve the desired result.

(a) To give the patient time to adapt to a particular medicine, the dosage is slowly increased over time. The patient is initially given a dosage of 0.75 units. The dosage is increased by 0.5 units each day until a total of at least 10 units of medicine has been taken. Find the least number of days it takes to achieve this, and the dosage on that day. [4]

(b) The doctor prescribed a new medication to another patient. After n days, the total dosage of the new medicine administered to the patient is given by $\frac{3}{20}\left(\frac{5^n-4^n}{5^{n-1}}\right)$ units.

  (i) Show that the increase in dosage of the new medicine administered per day follows a geometric progression. [3]

  (ii) Find the total dosage of the new medicine administered to the patient after a long period of treatment. [2]

(c) Doctors can manage the medication so that the daily dosage of medicine increases as an arithmetic progression. Explain in context why the arithmetic progression is not a preferred model for the dosage of medication. [1]

8 Differentiation

8.1 Basic Differentials

y	$\frac{\mathrm{d}y}{\mathrm{d}x}$
x^n	nx^{n-1}
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\log_b x$	$\frac{1}{x\ln b}$, b constant
$\mathrm{e}^{\mathrm{f}(x)}$	$\mathrm{f}'(x)\,\mathrm{e}^{\mathrm{f}(x)}$
$a^{\mathrm{f}(x)}$	$\mathrm{f}'(x)\,a^{\mathrm{f}(x)}\ln a$, a constant
$\mathrm{p}(x)\,\mathrm{q}(x)$	$\mathrm{p}'(x)\,\mathrm{q}(x)+\mathrm{p}(x)\,\mathrm{q}'(x)$
$\frac{\mathrm{p}(x)}{\mathrm{q}(x)}$	$\frac{\mathrm{p}'(x)\,\mathrm{q}(x)-\mathrm{p}(x)\,\mathrm{q}'(x)}{[\mathrm{q}(x)]^2}$

y	$\frac{\mathrm{d}y}{\mathrm{d}x}$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\frac{1}{1+x^2}$
$\sec x$	$\sec x\tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x\cot x$
$\cot x$	$-\operatorname{cosec}^2 x$

Important Note

i Remember to multiply by the differential when you are differentiating. Bracket, not divide.

8.1.1 Implicit Differentiation

Example 8.1

$$x^3y+1+\ln y=\cos(xy)$$

Differentiate both sides with respect to x:

$$\begin{aligned}
3x^2y+x^3\frac{\mathrm{d}y}{\mathrm{d}x}+0+\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x}&=-\sin(xy)\left(y+x\frac{\mathrm{d}y}{\mathrm{d}x}\right)\\
3x^2y+x^3\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x}&=-y\sin(xy)-x\sin(xy)\frac{\mathrm{d}y}{\mathrm{d}x}\\
x^3\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x}+x\sin(xy)\frac{\mathrm{d}y}{\mathrm{d}x}&=-y\sin(xy)-3x^2y\\
\frac{\mathrm{d}y}{\mathrm{d}x}\left(x^3+\frac{1}{y}+x\sin(xy)\right)&=-y\left(\sin(xy)+3x^2\right)\\
\frac{\mathrm{d}y}{\mathrm{d}x}\left(\frac{x^3y+1+xy\sin(xy)}{y}\right)&=-y\left(\sin(xy)+3x^2\right)\\
\therefore\ \frac{\mathrm{d}y}{\mathrm{d}x}&=-\frac{y^2\left(\sin(xy)+3x^2\right)}{x^3y+1+xy\sin(xy)}
\end{aligned}$$

8.1.2 Parametric Differentiation

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}y}{\mathrm{d}t}\times\frac{\mathrm{d}t}{\mathrm{d}x}$$

Example 8.2

$$x=t+\sin t,\quad y=1-\cos t$$

$$\begin{aligned}
&\frac{\mathrm{d}x}{\mathrm{d}t}=1+\cos t,\quad \frac{\mathrm{d}y}{\mathrm{d}t}=\sin t\\
&\frac{\mathrm{d}t}{\mathrm{d}x}=\frac{1}{\frac{\mathrm{d}x}{\mathrm{d}t}}=\frac{1}{1+\cos t}\\
&\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}y}{\mathrm{d}t}\times\frac{\mathrm{d}t}{\mathrm{d}x}\\
&\phantom{\frac{\mathrm{d}y}{\mathrm{d}x}}=\frac{\sin t}{1+\cos t}\\
&\phantom{\frac{\mathrm{d}y}{\mathrm{d}x}}=\frac{2\sin\frac{t}{2}\cos\frac{t}{2}}{2\cos^2\frac{t}{2}}\\
&\phantom{\frac{\mathrm{d}y}{\mathrm{d}x}}=\frac{\sin\frac{t}{2}}{\cos\frac{t}{2}}\\
\therefore\ &\frac{\mathrm{d}y}{\mathrm{d}x}=\tan\frac{t}{2}
\end{aligned}$$

9 Applications of Differentiation

9.1 Tangents & Normals

A tangent can be diagonal, vertical or horizontal.

Gradient of tangent = value of $\frac{\mathrm{d}y}{\mathrm{d}x}$.

Gradient of normal:

$$m_{\text{tangent}} \times m_{\text{normal}} = -1$$

Equation of tangent: $y = mx + c$. To find c, substitute any point on the tangent.

9.1.1 Standard Curves

Example 9.1

Find the equation of the tangent to $y = x^2 - 3x + 5$ at $x = 2$.

$$\begin{aligned}
\text{When } x = 2, \quad & y = 2^2 - 3(2) + 5 = 3 \\
& \frac{\mathrm{d}y}{\mathrm{d}x} = 2x - 3 \\
\text{When } x = 2, \quad & \frac{\mathrm{d}y}{\mathrm{d}x} = 2(2) - 3 = 1 \\
& y = x + c \\
& 3 = 2 + c \\
& c = 1 \\
& \therefore y = x + 1
\end{aligned}$$

9.1.2 Implicit Curves

Example 9.2

[EJC 2025 Prelim P2 Q1(b)]

The curve C has equation $x^3 + y^3 - 3xy = k$, where k is a positive constant. There are two points on C where the tangents are parallel to the y-axis. Given that $k = 3$, find the exact y-coordinates of these two points.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y - x^2}{y^2 - x}$$

Tangent parallel to y-axis ⇒ $\frac{\mathrm{d}y}{\mathrm{d}x}$ undefined:

$$\begin{aligned} y^2-x&=0 \\ x&=y^2 \\ y^6-2y^3-3&=0 \\ (y^3-3)(y^3+1)&=0 \\ \therefore y&=3^{\frac{1}{3}} \quad \text{or} \quad y=-1 \end{aligned}$$

9.1.3 Parametric Curves

Example 9.3

[MI 2025 Prelim P2 Q2(i)]

A curve C has parametric equations $x=\sqrt{t}$, $y=t^2+4$, where $0\leqslant t\leqslant 16$. Find the equation of the tangent to C at the point P with parameter p.

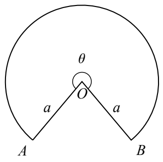
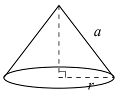
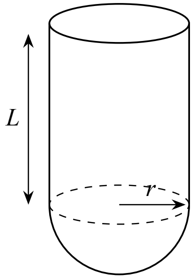
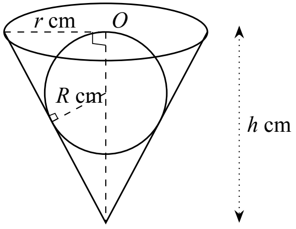
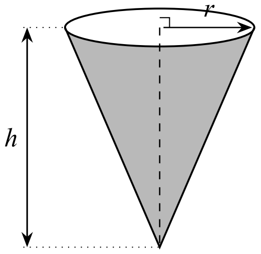
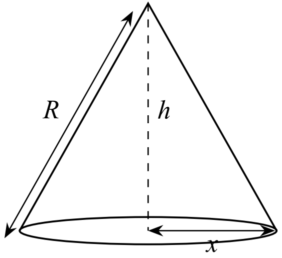
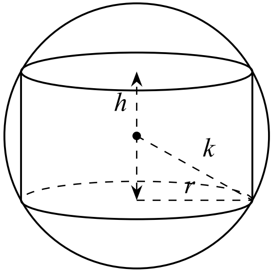
$$\begin{aligned} \frac{\mathrm{d}x}{\mathrm{d}t}&=\frac{1}{2\sqrt{t}}, \quad \frac{\mathrm{d}y}{\mathrm{d}t}=2t \\ \frac{\mathrm{d}y}{\mathrm{d}x}&=\frac{2t}{\frac{1}{2\sqrt{t}}}=4t^{\frac{3}{2}} \\ y-(p^2+4)&=4p^{\frac{3}{2}}\left(x-\sqrt{p}\right) \\ \therefore y&=4p^{\frac{3}{2}}x-3p^2+4 \end{aligned}$$

9.2 Max/Min

- To find the stationary value of y, solve $\frac{\mathrm{d}y}{\mathrm{d}x}=0$.
- Then, to find/prove its nature, perform the first/second derivative test.

Important Note

In many questions, you may be asked to prove the equation of the quantity you are trying to maximise. It's important to not spend too long on it if you can't prove it. Just continue with the rest of the question to differentiate and solve $\frac{\mathrm{d}y}{\mathrm{d}x}=0$.

Index	Diagram	Info	Equation	Comments
1	[TJC 2025 Prelim P2 Q4]  Figure 1  Figure 2	Constant: radius a Variables: angle θ	Prove that $V^2=\frac{a^6}{576\pi^4}\theta^4\left(4\pi^2-\theta^2\right)$	Take advantage of the constant slant height and major arc length AB.
2	[DHS 2025 Prelim P1 Q11]  	Constant: budget \$$k$ Variables: r	Prove that $V=\frac{k}{20}r-\frac{10}{3}\pi r^3$	Use the constant budget k to eliminate L.
3	[TJC 2024 Promo Q12]  	Constant: sphere radius R Variables: h	Prove that $r^2=\frac{h^2R^2}{h^2-2hR}$	Take advantage of the constant sphere radius R, and similar triangles. Pythagoras theorem is useful here too.
4	[CJC 2024 Promo Q10]  	Constant: volume $20\pi\ \text{cm}^3$ Variables: r	Prove that $A\frac{\mathrm{d}A}{\mathrm{d}r}=\pi^2\left(2r^3-\frac{3600}{r^3}\right)$	Use the constant volume to eliminate h. For rate of change questions, use similar triangles.
5	[EJC 2018 Promo Q7]  	Constant: slant edge R Variables: x	Prove that $9V^2=\pi^2x^4\left(R^2-x^2\right)$	Use the constant R to eliminate h.
6	[TMJC 2024 Prelim P1 Q2]  	Constant: sphere radius k Variables: k, r	—	Use the constant radius of sphere k and Pythagoras theorem is useful here too.

9.3 Rate of Change

$$\frac{\mathrm{d}y}{\mathrm{d}t}=\frac{\mathrm{d}y}{\mathrm{d}x}\times\frac{\mathrm{d}x}{\mathrm{d}t}$$

Step 1: Identify what information is given.

Step 2: Form the missing equation if necessary and differentiate it.

Example 9.4

Given $y=x^2-4x+1$ and $\frac{\mathrm{d}x}{\mathrm{d}t}=3$, find the rate of change of y when $x=5$.

$$\frac{\mathrm{d}y}{\mathrm{d}x}=2x-4$$

$$\text{When } x=5,\quad \frac{\mathrm{d}y}{\mathrm{d}x}=2(5)-4=6$$

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}t}&=\frac{\mathrm{d}y}{\mathrm{d}x}\times\frac{\mathrm{d}x}{\mathrm{d}t}\\&=6\times3\\ \therefore \frac{\mathrm{d}y}{\mathrm{d}t}&=18\end{aligned}$$

10 Power Series

10.1 Standard Formulae (MF27)

$$(1+x)^n=1+nx+\frac{n(n-1)}{2!}x^2+\cdots+\frac{n(n-1)\cdots(n-r+1)}{r!}x^r+\ldots \quad (|x|<1) \tag{1}$$

$$e^x=1+x+\frac{x^2}{2!}+\cdots+\frac{x^r}{r!}+\ldots \quad (\text{all } x) \tag{2}$$

$$\sin x=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots+\frac{(-1)^r x^{2r+1}}{(2r+1)!}+\ldots \quad (\text{all } x) \tag{3}$$

$$\cos x=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\cdots+\frac{(-1)^r x^{2r}}{(2r)!}+\ldots \quad (\text{all } x) \tag{4}$$

$$\ln(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}-\cdots+\frac{(-1)^{r+1}x^r}{r}+\ldots \quad (-1<x\leqslant 1) \tag{5}$$

Important Note

i For equations (1) and (5), the constant must be 1.

CAUTION

i Any substituted expressions of x cannot have + constant inside.

✗ $$e^{1+x+x^2+\cdots}=1+(1+x+x^2+\cdots)+\frac{(1+x+x^2+\cdots)^2}{2!}+\cdots$$

✓ $$\begin{aligned} e^{1+x+x^2+\cdots}&=e^1\cdot e^{x+x^2+\cdots}\\ &=e\left[1+(x+x^2+\cdots)+\frac{(x+x^2+\cdots)^2}{2!}+\cdots\right]\end{aligned}$$

10.2 Maclaurin Series

$$f(x)=f(0)+x\,f'(0)+\frac{x^2}{2!}f''(0)+\cdots+\frac{x^r}{r!}f^{(r)}(0)+\ldots$$

Conditions for existence: for all $n\in\mathbb{Z}^+\cup\{0\}$, $f^{(n)}$ is differentiable and $f^{(n)}(0)$ is defined.

10.3 Small Angle Approximations

When θ is sufficiently small, θ^3 and above can be ignored.

$$\begin{aligned}\sin\theta &\approx \theta\\ \cos\theta &\approx 1-\frac{\theta^2}{2}\\ \tan\theta &\approx \theta\end{aligned}$$

Example 10.1

$$\sin 2\theta \approx 2\theta$$

Example 10.2

$$\begin{aligned}\cos 3\theta &\approx 1-\frac{(3\theta)^2}{2}\\ &=1-\frac{9\theta^2}{2}\end{aligned}$$

Example 10.3

$$\begin{aligned}\sec\theta &= \frac{1}{\cos\theta}\\ &\approx \left(1-\frac{\theta^2}{2}\right)^{-1}\\ &\approx 1+\frac{\theta^2}{2}\end{aligned}$$

Example 10.4

$\sin\left(\frac{\pi}{4}+\theta\right) \not\approx \frac{\pi}{4}+\theta$

$$\begin{aligned}\sin\left(\frac{\pi}{4}+\theta\right) &= \sin\frac{\pi}{4}\cos\theta+\cos\frac{\pi}{4}\sin\theta\\ &= \frac{1}{\sqrt{2}}(\cos\theta+\sin\theta)\\ &\approx \frac{1}{\sqrt{2}}\left(1-\frac{\theta^2}{2}+\theta\right)\\ \therefore\ \sin\left(\frac{\pi}{4}+\theta\right) &\approx \frac{1}{\sqrt{2}}\left(1+\theta-\frac{\theta^2}{2}\right)\end{aligned}$$

10.4 General Approximations

- Values of x closer to 0 give better approximations as higher powers of x are less significant.
- Values of x outside the range of validity give an inaccurate approximation.

11 Integration

11.1 Basic Integrals

Basic	**General form** (let a, b, k be constants)
$\int x^n \, \mathrm{d}x = \frac{x^{n+1}}{n+1} + c, \quad n \neq -1$	$\int (ax+b)^n \, \mathrm{d}x = \frac{(ax+b)^{n+1}}{a(n+1)} + c, \quad n \neq -1$
$\int \frac{1}{x} \, \mathrm{d}x = \ln \lvert x \rvert + c$	$\int \frac{1}{ax+b} \, \mathrm{d}x = \frac{1}{a} \ln \lvert ax+b \rvert + c$
$\int \mathrm{e}^x \, \mathrm{d}x = \mathrm{e}^x + c$	$\int \mathrm{e}^{kx} \, \mathrm{d}x = \frac{1}{k} \mathrm{e}^{kx} + c$
$\int \sin x \, \mathrm{d}x = -\cos x + c$	$\int \sin kx \, \mathrm{d}x = -\frac{1}{k} \cos kx + c$
$\int \cos x \, \mathrm{d}x = \sin x + c$	$\int \cos kx \, \mathrm{d}x = \frac{1}{k} \sin kx + c$
$\int \sec^2 x \, \mathrm{d}x = \tan x + c$	$\int \sec^2 kx \, \mathrm{d}x = \frac{1}{k} \tan kx + c$

11.2 $\mathrm{f}'(x)$ Formulas

$$\int \mathrm{f}'(x) \, [\mathrm{f}(x)]^n \, \mathrm{d}x = \frac{[\mathrm{f}(x)]^{n+1}}{n+1} + c, \qquad n \neq -1$$

$$\int \frac{\mathrm{f}'(x)}{\mathrm{f}(x)} \, \mathrm{d}x = \ln \lvert \mathrm{f}(x) \rvert + c$$

$$\int \mathrm{f}'(x) \, \mathrm{e}^{\mathrm{f}(x)} \, \mathrm{d}x = \mathrm{e}^{\mathrm{f}(x)} + c$$

11.3 New Trigonometric Integrals

- To find $\int \tan x \, \mathrm{d}x$, $\int \sec x \, \mathrm{d}x$, $\int \operatorname{cosec} x \, \mathrm{d}x$, $\int \cot x \, \mathrm{d}x$ — see the **MF27**.
- To find $\int \sin^2 x \, \mathrm{d}x$ or $\int \cos^2 x \, \mathrm{d}x$, use the **double angle formula** for $\cos 2\theta$.
  To find $\int \tan^2 x \, \mathrm{d}x$, use the identity $\tan^2 x = \sec^2 x - 1$.
- The differentiation formulas for $\sec x$ and $\operatorname{cosec} x$ in the MF27 can also be used as **integration** formulas. In particular, the missing formula yields

$$\int \operatorname{cosec}^2 x \, \mathrm{d}x = -\cot x + c.$$

11.4 Standard MF27 Formulas

Standard	General form
$\int \frac{1}{a^2+x^2}\,dx=\frac{1}{a}\tan^{-1}\frac{x}{a}+c$	$\int \frac{f'(x)}{a^2+[f(x)]^2}\,dx=\frac{1}{a}\tan^{-1}\frac{f(x)}{a}+c$
$\int \frac{1}{\sqrt{a^2-x^2}}\,dx=\sin^{-1}\frac{x}{a}+c$	$\int \frac{f'(x)}{\sqrt{a^2-[f(x)]^2}}\,dx=\sin^{-1}\frac{f(x)}{a}+c$
$\int \frac{1}{x^2-a^2}\,dx=\frac{1}{2a}\ln\left\lvert\frac{x-a}{x+a}\right\rvert+c$	$\int \frac{f'(x)}{[f(x)]^2-a^2}\,dx=\frac{1}{2a}\ln\left\lvert\frac{f(x)-a}{f(x)+a}\right\rvert+c$
$\int \frac{1}{a^2-x^2}\,dx=\frac{1}{2a}\ln\left\lvert\frac{a+x}{a-x}\right\rvert+c$	$\int \frac{f'(x)}{a^2-[f(x)]^2}\,dx=\frac{1}{2a}\ln\left\lvert\frac{a+f(x)}{a-f(x)}\right\rvert+c$

IMPORTANT

a must be a (positive) **constant**.

Important Remark

When using these formulas, it is **VERY COMMON** to have to complete the square first.

Remark

These formulas are commonly used together with $f'(x)$ formulas to integrate **rational functions**.

Example 11.1

$$\begin{aligned}\int \frac{4x}{\sqrt{4x-x^2}}\,dx&=\int \frac{-2(-2x+4)+8}{\sqrt{4x-x^2}}\,dx\\&=-2\int(-2x+4)\,(4x-x^2)^{-\frac{1}{2}}\,dx+8\int\frac{1}{\sqrt{4-(x-2)^2}}\,dx\\&=-2\cdot\frac{(4x-x^2)^{\frac{1}{2}}}{\left(\frac{1}{2}\right)}+8\sin^{-1}\frac{x-2}{2}+c\\&=-4\sqrt{4x-x^2}+8\sin^{-1}\frac{x-2}{2}+c\end{aligned}$$

11.5 Integration by Parts (IKID)

Integrate Keep Integrate Differentiate

$$IK-\int ID\,\mathrm{d}x$$

Example 11.2

$$\int \underbrace{x}_{\text{K/D}}\ \underbrace{\mathrm{e}^x}_{\text{I}}\,\mathrm{d}x=\underbrace{\mathrm{e}^x}_{\text{I}}\ \underbrace{x}_{\text{K}}-\int \underbrace{\mathrm{e}^x}_{\text{I}}\ \underbrace{1}_{\text{D}}\,\mathrm{d}x$$

- It is possible to make the **wrong choice** of which expression to integrate. In such scenarios, the $\int ID\,\mathrm{d}x$ will look *worse* than the original problem.
- It is common to have to **IKID twice**. Consider this approach if you are unable to continue.

IMPORTANT

Integration by parts is often used as the **last resort**, i.e. please consider **ALL OTHER INTEGRATION TECHNIQUES** first.

Common tricks

- **IKID by multiplying by 1.**

$$\int \ln x\,\mathrm{d}x=\int 1\cdot\ln x\,\mathrm{d}x=\ldots$$

- **Non-obvious splitting.**

$$\int x^3\mathrm{e}^{x^2}\,\mathrm{d}x=\int x^2\cdot x\mathrm{e}^{x^2}\,\mathrm{d}x=\ldots$$

Here, we can integrate $x\mathrm{e}^{x^2}$ using the $\int \mathrm{f}'(x)\mathrm{e}^{\mathrm{f}(x)}\,\mathrm{d}x$ formula.

- **Loop method.**

$$\begin{aligned}\int \mathrm{e}^x\cos x\,\mathrm{d}x&=\mathrm{e}^x\cos x+\int \mathrm{e}^x\sin x\,\mathrm{d}x\\&=\mathrm{e}^x\cos x+\mathrm{e}^x\sin x-\int \mathrm{e}^x\cos x\,\mathrm{d}x\end{aligned}$$

Therefore,

$$\begin{aligned}2\int \mathrm{e}^x\cos x\,\mathrm{d}x&=\mathrm{e}^x\cos x+\mathrm{e}^x\sin x+c\\\int \mathrm{e}^x\cos x\,\mathrm{d}x&=\frac{1}{2}\left(\mathrm{e}^x\cos x+\mathrm{e}^x\sin x\right)+c\end{aligned}$$

11.6 Integration by Substitution

Given $\int \mathrm{f}(x)\,\mathrm{d}x$ and a substitution $u = \dots$, apply:

Step 1: Substitute u into $\mathrm{f}(x)$.

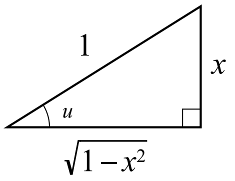
Step 2: Find $\frac{\mathrm{d}x}{\mathrm{d}u}$, and substitute it into the integral.

Step 3: *(Only for definite integrals)* Substitute the limits of integration from x values to u values.

Indefinite Integral

Example 11.3

$$\begin{aligned}
&\int \sqrt{1-x^2}\,\mathrm{d}x, \qquad x=\sin u\\
&=\int \underbrace{\sqrt{1-\sin^2 u}}_{\text{Step 1}}\ \underbrace{\cos u\,\mathrm{d}u}_{\text{Step 2}}\\
&=\int \sqrt{\cos^2 u}\ \cos u\,\mathrm{d}u\\
&=\int \cos^2 u\,\mathrm{d}u\\
&=\frac{1}{2}\int (1+\cos 2u)\,\mathrm{d}u\\
&=\frac{1}{2}\left(u+\frac{1}{2}\sin 2u\right)+c\\
&=\frac{1}{2}u+\frac{1}{2}\sin u\cos u+c\\
&=\frac{1}{2}\sin^{-1}x+\frac{1}{2}x\sqrt{1-x^2}+c
\end{aligned}$$



$\sin u = x$

$\cos u = \sqrt{1-x^2}$

Definite Integral

Example 11.4

$$\int_0^{\frac{1}{2}} \sqrt{1-x^2}\,\mathrm{d}x, \qquad x=\sin u$$

$$x=0 \Rightarrow u=0$$

$$x=\frac{1}{2} \Rightarrow u=\frac{\pi}{6}$$

$$\begin{aligned}
\int_0^{\frac{1}{2}} \sqrt{1-x^2}\,\mathrm{d}x &= \int_0^{\frac{\pi}{6}} \sqrt{1-\sin^2 u}\ \cos u\,\mathrm{d}u\\
&=\int_0^{\frac{\pi}{6}} \cos^2 u\,\mathrm{d}u\\
&=\frac{1}{2}\int_0^{\frac{\pi}{6}} (1+\cos 2u)\,\mathrm{d}u\\
&=\frac{1}{2}\left[u+\frac{1}{2}\sin 2u\right]_0^{\frac{\pi}{6}}\\
&=\frac{1}{2}\left[\left(\frac{\pi}{6}+\frac{1}{2}\sin\frac{\pi}{3}\right)-\left(0+\frac{1}{2}\sin 0\right)\right]\\
&=\frac{1}{2}\left(\frac{\pi}{6}+\frac{\sqrt{3}}{4}\right)\\
&=\frac{\pi}{12}+\frac{\sqrt{3}}{8}
\end{aligned}$$

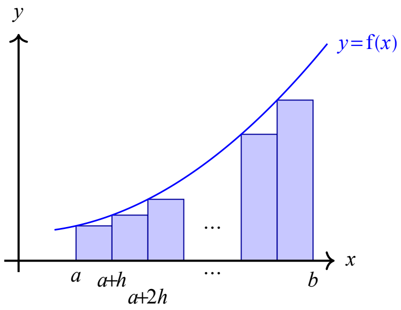
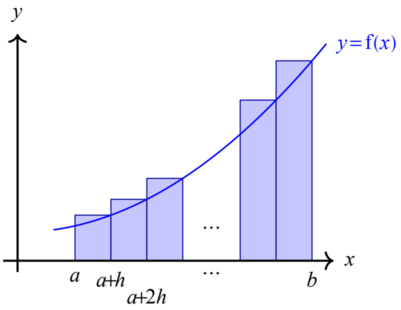
Remark

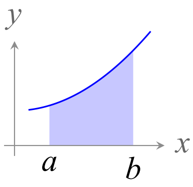
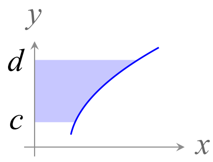
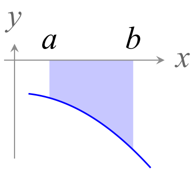
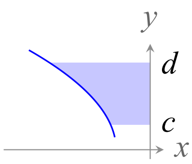
i Remember to sub the final answer back to be in terms of x, unless it's a **definite integral**.

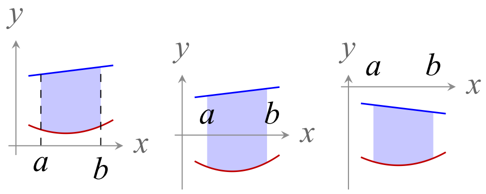
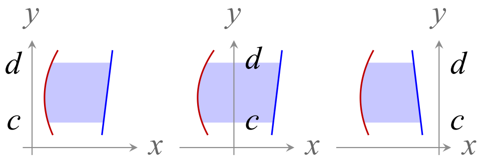
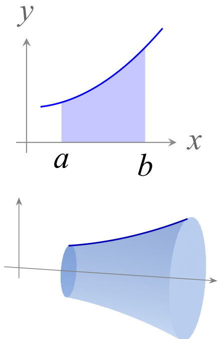
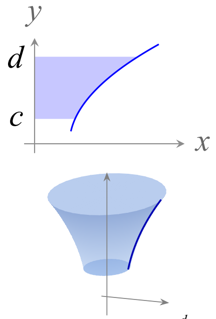
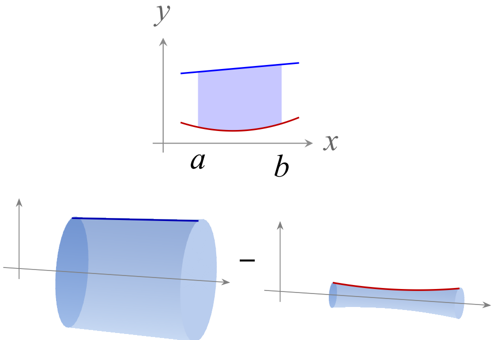
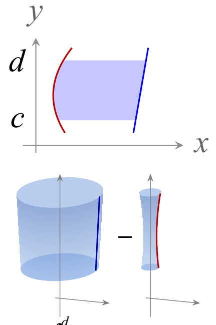
12 Definite Integrals

$$\int_a^b \mathrm{f}(x)\,\mathrm{d}x=\Big[\mathrm{F}(x)\Big]_a^b=\mathrm{F}(b)-\mathrm{F}(a)$$

12.1 Area as a Limit of a Sum

Left-hand sum	**Right-hand sum**
	
$\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n-1}\frac{b-a}{n}\,\mathrm{f}\left(a+\frac{r(b-a)}{n}\right)=\int_a^b \mathrm{f}(x)\,\mathrm{d}x$	$\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\frac{b-a}{n}\,\mathrm{f}\left(a+\frac{r(b-a)}{n}\right)=\int_a^b \mathrm{f}(x)\,\mathrm{d}x$
Increasing $\mathrm{f}(x)$ ⇒ underestimate Decreasing $\mathrm{f}(x)$ ⇒ overestimate	Increasing $\mathrm{f}(x)$ ⇒ overestimate Decreasing $\mathrm{f}(x)$ ⇒ underestimate

Index	dx	dy
Area under curve	 $\int_a^b y\,\mathrm{d}x$	 $\int_c^d x\,\mathrm{d}y$
Negative areas	 $-\int_a^b y\,\mathrm{d}x$	 $-\int_c^d x\,\mathrm{d}y$

Index	dx	dy
Areas between two graphs	 All 3 cases: $\int_a^b (y_1 - y_2)\ \mathrm{d}x$ y_1 (blue) above y_2 (red)	 All 3 cases: $\int_c^d (x_1 - x_2)\ \mathrm{d}y$ x_1 (blue) right of x_2 (red)
Volume rotated about axis	 About the x-axis: $\pi\int_a^b y^2\ \mathrm{d}x$	 About the y-axis: $\pi\int_c^d x^2\ \mathrm{d}y$
Volume generated by region between two curves	 $\pi\int_a^b (y_1^2 - y_2^2)\ \mathrm{d}x$	 $\pi\int_c^d (x_1^2 - x_2^2)\ \mathrm{d}y$

When finding **area**, we generally have the choice to integrate via dx or dy. Some considerations:

- dx or dy easier to integrate
- do we need to add or subtract regular shapes like rectangles
- is it even possible to make x (or y) the subject?

For **volume**, we cannot choose which variable to integrate with respect to.

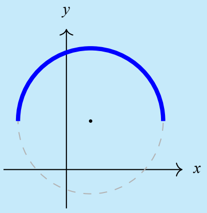
- Region rotated about the x-axis: $\pi\int_a^b y^2\,\mathrm{d}x$
- Region rotated about the y-axis: $\pi\int_c^d x^2\,\mathrm{d}y$

Important Note: dealing with conics

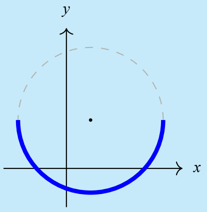
When making y the subject, the sign of the square root indicates which half of the conic is being considered.

Example $(x-1)^2+(y-2)^2=9$

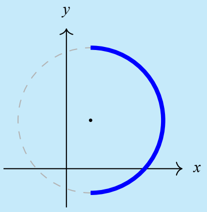
$$y=2+\sqrt{9-(x-1)^2}$$



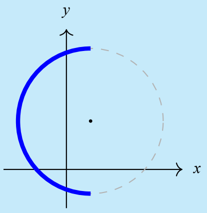
$$y=2-\sqrt{9-(x-1)^2}$$



$$x=1+\sqrt{9-(y-2)^2}$$



$$x=1-\sqrt{9-(y-2)^2}$$



Note: this concept can be used for any conic. That is, circles, ellipses, hyperbolas, parabolas.

Notes on volume

- "Rotate π radians about …" does not usually mean the formula has to be divided by 2. It just means that rotating only 180° is sufficient to generate the full solid. The volume remains as $\pi\int_a^b y^2\,\mathrm{d}x$.
- To find the volume of rotating regions "below" the axis, there is no need to put a negative sign in front:

$$✗ \quad -\pi\int_a^b y^2\,\mathrm{d}x \qquad\qquad ✓ \quad \pi\int_a^b y^2\,\mathrm{d}x$$

- Common mistake for volume between two graphs:

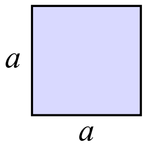
$$✗ \quad \pi\int_a^b (y_1-y_2)^2 \, dx \qquad\qquad ✓ \quad \pi\int_a^b (y_1^2-y_2^2) \, dx$$

> **Important Note**
>
> Take note if the question asks for exact or not. If they did not, please evaluate the integral using GC rather than integration by hand.

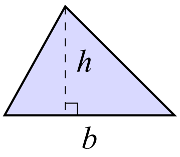
> **Remark**
>
> If an integral seems very difficult/impossible to integrate, then check if the question wants exact form OR check if you have integrated something similar in earlier parts.

Regular Shapes in Area and Volume

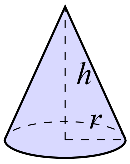
Sometimes we may encounter familiar shapes like squares, rectangles, triangles, trapeziums, cones, and cylinders. It will be useful to use the formulas for these shapes/solids for efficiency.



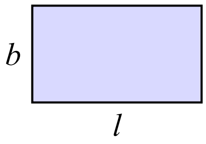
Square: Area $= a^2$



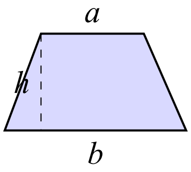
Triangle: Area $= \frac{1}{2}bh$



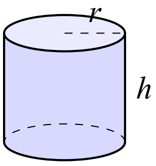
Cone: Volume $= \frac{1}{3}\pi r^2 h$



Rectangle: Area $= lb$



Trapezium: Area $= \frac{1}{2}(a+b)h$



Cylinder: Volume $= \pi r^2 h$

Integration with Modulus

To integrate with modulus, we determine where the function inside the modulus is positive and where it is negative, then split it up.

Example 12.1

$$\begin{aligned}\int_0^{\pi} |x\sin x\cos x| \, dx &= \int_0^{\frac{\pi}{2}} (x\sin x\cos x) \, dx + \int_{\frac{\pi}{2}}^{\pi} -(x\sin x\cos x) \, dx \\ &= \int_0^{\frac{\pi}{2}} \frac{1}{2} x\sin 2x \, dx + \int_{\frac{\pi}{2}}^{\pi} -\frac{1}{2} x\sin 2x \, dx \\ &= \dots\end{aligned}$$

13 Differential Equations

13.1 Definition of a Differential Equation

Differential Equation

A **differential equation** is an equation involving a derivative, e.g. $\frac{dy}{dx}$, of an unknown function.

13.2 Solving a Differential Equation

Solving a differential equation means finding y as a function of x that satisfies the differential equation.

General and Particular Solutions

- **General solution**: a solution containing arbitrary constant(s).
- **Particular solution**: a solution where the constant(s) are found from given conditions.

In this chapter, C and D denote arbitrary constants that are added, and A denotes an arbitrary constant that multiplies.

13.2.1 $\frac{dy}{dx} = f(x)$

Example 13.1	**Example 13.2**	**Example 13.3**
$\frac{dy}{dx}=3x^2$	$\frac{dy}{dx}=e^{2x}$	$\frac{d^2y}{dx^2}=6x$
$\vdots$	$\vdots$	$\vdots$
$y=x^3+C$	$y=\frac{1}{2}e^{2x}+C$	$y=x^3+Cx+D$

13.2.2 Method of Separable Variables

When a differential equation has "y" terms, we apply the method of separable variable. Divide the expression in terms of y to the LHS, and keep all x terms on the RHS.

$$\boxed{f(y)\frac{dy}{dx}=g(x)}$$

Example 13.4

$$\frac{\mathrm{d}y}{\mathrm{d}x}=k(30-y)$$

$$\frac{1}{30-y}\frac{\mathrm{d}y}{\mathrm{d}x}=k$$

$$\vdots$$

$$\therefore y=30-Ae^{-kx}$$

Example 13.5

$$\frac{\mathrm{d}y}{\mathrm{d}x}=ky(1-y)$$

$$\frac{1}{y(1-y)}\frac{\mathrm{d}y}{\mathrm{d}x}=k$$

$$\vdots$$

$$\therefore y=\frac{1}{1+Ae^{-kx}}$$

Example 13.6

$$\frac{\mathrm{d}y}{\mathrm{d}x}=xy^2+x+y^2+1$$

$$=x\left(y^2+1\right)+\left(y^2+1\right)$$

$$=(x+1)\left(y^2+1\right)$$

$$\frac{1}{y^2+1}\frac{\mathrm{d}y}{\mathrm{d}x}=x+1$$

$$\vdots$$

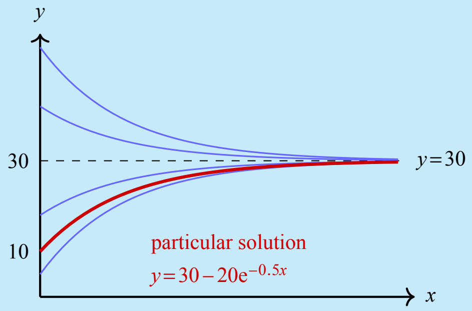
$$\therefore y=\tan\left(\frac{1}{2}x^2+x+C\right)$$

Note

If some values of x and y are given (usually initial values $x=0$, $y=y_0$), you should solve for the particular solution.

Note: Sketching graphs of solutions

- **General solution**: family of curves, i.e. many possible different curves.
- **Particular solution**: one curve.



Family $y=30-Ae^{-0.5x}$

Note

Most of the time, graphs of solutions to differential equations in H2 Math have horizontal asymptotes. We can find them by letting $x\to\infty$. Alternatively, let $\frac{\mathrm{d}y}{\mathrm{d}x}=0$ and solve for y if it is a separable variable differential equation.

13.3 Substitution

Given a differential equation, it may sometimes be impossible to separate the variables. In such cases, a substitution will be given to you (do not come up with your own).

Example 13.7

(DHS 2025 Prelim P1 Q1) It is given that $2xy\frac{dy}{dx}=x^2-y^2$ where $x>0$, $y<0$. Using the substitution $u=\frac{y}{x}$, show that the differential equation can be transformed to $\frac{2u}{1-3u^2}\frac{du}{dx}=\frac{1}{x}$. Hence find the general solution of y in terms of x.

Step 1: Differentiate the substitution

$$u=\frac{y}{x} \Rightarrow y=ux$$

$$\frac{dy}{dx}=\frac{du}{dx}x+u \qquad \dots(1)$$

Step 2: Substitute the substitution

$$2xy\frac{dy}{dx}=x^2-y^2$$

$$2x(ux)\left(x\frac{du}{dx}+u\right)=x^2-(ux)^2$$

$$\vdots$$

$$\frac{2u}{1-3u^2}\frac{du}{dx}=\frac{1}{x} \quad \text{(shown)}$$

$$\vdots$$

$$3u^2=1-\frac{A}{x^3}$$

$$\left(\frac{y}{x}\right)^2=\frac{1}{3}\left(1-\frac{A}{x^3}\right)$$

$$\therefore y=-\sqrt{\frac{x^2}{3}\left(1-\frac{A}{x^3}\right)}=-x\sqrt{\frac{1}{3}\left(1-\frac{A}{x^3}\right)} \qquad (\because y<0)$$

!

VERY IMPORTANT NOTE

For substitution in differential equations, always remember to substitute the original variables (x and y) back into the equation!

!

SUPER ULTRA IMPORTANT NOTE

The variable in the substitution is **NOT A CONSTANT** unless otherwise stated!

Example: $y=ux^2$ (product rule: u and x^2 are both variables)

✗ $\frac{dy}{dx}=2ux$ ✓ $\frac{dy}{dx}=x^2\frac{du}{dx}+u(2x)=x^2\frac{du}{dx}+2ux$

14 Complex Numbers

14.1 Cartesian Form

Definition 14.1.1. Cartesian form of a complex number

Every complex number $z \in \mathbb{C}$ can be expressed as $z = x + y\mathrm{i}$, $x, y \in \mathbb{R}$.

- A complex number z is **real** if $\operatorname{Im}(z) = 0$.
- A complex number z is **purely imaginary** if $\operatorname{Re}(z) = 0$.

14.2 Complex Number Arithmetic

- **Addition & subtraction**: similar to surds.
- **Multiplication**: similar to surds.
- **Division**: similar to surds, remember to *realise*.
- **Square roots**

Example 14.1

Find the values of $\sqrt{3+4\mathrm{i}}$.

$$\begin{aligned}(a+b\mathrm{i})^2 &= 3+4\mathrm{i}\\ a^2-b^2+2ab\,\mathrm{i} &= 3+4\mathrm{i}\\ a^2-b^2 &= 3\\ 2ab &= 4\\ a=\pm 2,\ b &= \pm 1\\ \therefore \sqrt{3+4\mathrm{i}} &= \pm(2+\mathrm{i})\end{aligned}$$

14.2.1 Simultaneous Equations

Common types:

1. Standard z, w (no conjugate, no modulus, no square)

$$\begin{aligned}2z+w &= 2-4\mathrm{i}\\ \mathrm{i}z-w &= 2\mathrm{i}\end{aligned}$$

[TJC 2025 Prelim P1 Q8(a)]

2. With z^* and w (or vice versa, or both)

$$z+2z^*+w=6+\mathrm{i} \quad \text{and} \quad \mathrm{i}z-w=3$$

[MI 2025 Prelim P2 Q3(a)]

If $z=2+\mathrm{i}w$, then $z^*=2-\mathrm{i}w^*$.

3. With z^2 and w

$$v^2-\mathrm{i}w+2=0$$

$$v=w+\mathrm{i}+3$$

[NYJC 2024 Prelim P1 Q1]

The quadratic formula is useful here; avoid letting $z=a+b\mathrm{i}$ and expanding, though it works.

4. With z and $|w|$

$$z+w^*+5\mathrm{i}=10 \quad \text{and} \quad |w|^2=z+18+\mathrm{i}$$

[CJC 2017 Prelim P2 Q4(a)]

For $w=a+b\mathrm{i}$, $|w|=\sqrt{a^2+b^2}$.

Note

Please check your answers using your GC for **BOTH** equations! The answers are usually quite nice and not ugly numbers.

14.2.2 Polynomials

Complex Conjugate Root Theorem

If all the coefficients of a polynomial equation $\mathrm{P}(z)=0$ are real, and $z=\alpha$ is a root, then its conjugate $z=\alpha^*$ is also a root.